SPECIFICATION V0.54 · REVISION 0

LEBRE v0.54 — Specification and Technical Report

Online forecasting with statistically tested structural changes — exit included —, a precision expert that sleeps when unused and a trivial-reference starting point, under a declared computational budget

LEBRE: Lifecycle-governed Evidence-Based Resource Evolution
Architecture: structural core (memory + structural expert + dynamic averaging, with changes tested by e-processes, triggered removal and equivalence report) + M1 (precision expert, dormant when unused) + M2 (trivial reference)

STATUS
research version promoted by a pre-registered rule (2026-10-13), with errata 01; scope: one-step-ahead forecasting with inputs

FINAL EVALUATION
93 reserved series; no harm per family against the previous version (error ratio 0.989 to 1.000); M1 + M2 over the core: 0.955 in the previous evaluation

COST
median of 870 operations per step; 0.905 times the previous version's per series [95% CI 0.857; 0.932]; estimated memory 83.3 KB

SCOPE OF THE EVIDENCE
lower cost where M1 does not help and equal where it does; removal verified in development and validations; core evaluated in the budget class on hydrology and buildings

Contents

1. Executive summary §1
2. The problem, the premise and the scope §2
3. Design principles §3
4. Architecture overview §4
5. Theoretical basis of the structural core §5
6. Theoretical basis of M1 and M2: aggregation of forecasters §6
7. Lifecycle of a structural change §7
8. The condition of use: reference misadjustment §8
9. M1: precision expert and dormancy §9
10. M2: starting from the trivial reference §10
11. Output safeguards and interval §11
12. Computational cost and memory §12
13. Observability and explainability §13
14. Evaluation methodology §14
15. Evidence for the structural core §15
16. The core on a simulated microcontroller §16
17. Development of the layers and of v0.54 §17
18. Validations on new data §18
19. Pre-registered final evaluation, audit and errata §19
20. Evolution of the architecture and ablations §20
21. Position in the literature §21
22. Limits and claims we do not make §22
23. Parameters §23
24. Pseudocode and reproducibility conditions §24
25. References §25
26. Revision history §26

1. Executive summary

LEBRE is an **online** forecaster for time series with inputs: at every step it receives the inputs and, when available, the target value; it predicts the next value, learns from the error and moves on, without re-processing the past and without prior training. The per-step cost is counted operation by operation: over the 93 series of the final evaluation, a median of 870 floating-point operations per step.

It has three layers (§4). The **structural core** is a simple live model (a memory of the target itself and a linear regression on the inputs) whose structure only changes by experiment: a challenger in shadow proposes to add, remove, swap or refine the response to a whole input, and the change is only accepted when an *e-process* — a sequential test valid at any time — accrues enough evidence, within an online multiple-testing procedure. **M1** combines the core forecast with a precision expert (a recursive regression on the target's past and on lag-band means of the inputs). **M2** makes the forecast start from a declared trivial reference (zero, persistence or seasonal naive) and trust the adaptive part only when it proves better.

This version (v0.54) extends the principle "change only with evidence, at a known cost" to two things: **cost**, which is now spent only where there is use — M1 *sleeps* while its weight is negligible (§9) and the report of equivalent inputs becomes cheap (§12, §13) —, and the **exit** of structure — a removal test with its own evidence opens when a unit's contribution drops (§5.9, §7). None of these parts is new in isolation; the document describes the combination, the conditions under which it works and what was measured.

Premise: LEBRE **does not aim to be the most accurate forecaster**. The goal is to deliver useful forecasts, without severe failures, at a low and predictable per-step cost, without per-series tuning and with every structural change logged and justified. Expensive models are treated as **reference ceilings**, not as opponents.

What was measured

- **Pre-registered final evaluation of v0.54** (93 series reserved before any code of this version; basins, buildings, wind and solar plants and price and rate levels): no harm in any family against the previous version (error 0.989 to 1.000 times its own per family) and a median per-series cost of **0.905** times its own [95% CI 0.857; 0.932] (§19).
- **M1 and M2 on top of the core** (previous pre-registered final evaluation, 104 series): error 0.955 [0.951; 0.960] times that of the core alone, with no harm in any family; the gain comes from Basins (CAMELS-BR) 0.879, FX: levels 0.952, FX: returns 0.966, Buildings (BDG2) 0.993 (§19.4).
- **Reference ceilings:** against Chronos-2 with covariates (a foundation model of ~10^9 operations per forecast), a ratio of 0.84 to 1.34 per family in the final evaluation — below 1 in basins and levels, above in buildings, solar and wind (§19).
- **The core in its budget class** (its own pre-registered evaluation, 60 basin and building series): error relative to online NLinear of 0.812 [0.752; 0.866], the lowest among the eight comparators of similar cost evaluated, with no series with a severe failure (§15).
- **Stress audit** after the promotion (10 tests, from identity to 200 thousand steps): no correctness defect nor catastrophic failure; two defects found — one fixed in errata 01 (cost peak), the other declared as a limit (relations that disappear and return) (§19.5).

Main limits: one-step forecasting; removing a relation that disappears for more than ~5,200 steps and returns costs 5% to 10% of error on series with that pattern; spurious inputs in level series are still accepted; the cycle must be declared; the core's testing mechanism **did not improve accuracy** compared to switching all inputs on without tests (ratio 0.988 [0.970; 1.006] on 100 series, exploratory); there is no formal error guarantee in the original scale for M1 and M2; the microcontroller was simulated and only for the core. No component is original (§21, §22).

2. The problem, the premise and the scope

The problem

Many systems continuously measure a target (river flow, building consumption, plant generation, electric load, an exchange rate) together with inputs that influence it, often with delay (rain, temperature, upstream flows, other currencies). Forecasting the next value *on the device itself* requires three things at once: very low per-step cost, operation without per-series tuning or prior training, and a behaviour that does not fail severely when the stream has gaps, changes or is nearly unpredictable. The most accurate modern models (foundation models) cost 10^7 to 10^{10} operations per forecast; classical models fitted series by series need maximum-likelihood optimisation; and cheap online linear models do not know, by themselves, which inputs and delays matter, nor when it is better to risk nothing beyond a trivial forecast.

The premise

LEBRE occupies a specific point of the accuracy–cost–transparency trade-off. It is evaluated **within its budget class** (models with up to ~10^4 operations per forecast), with the expensive models as ceilings. "Delivering well" means: competitive error within the class, **no catastrophic failure**, not losing to a declared trivial forecast, predictable cost and an explanation of every change.

Scope

- **one-step-ahead** forecasting, online, with the current inputs available (as in control and very-short-term forecasting);
- series with **few inputs** and an approximately linear relation, with delays of up to 31 steps in the core and up to 15 in the precision expert;
- the environment only provides the sampling period (e.g. 24 for hourly data with a daily cycle; for buildings, also 168 for the weekly cycle) and the trivial reference suited to the target (zero for changes such as financial returns, seasonal naive when there is a cycle, persistence otherwise);
- inputs standardised causally (exponential mean and variance), as an input contract.

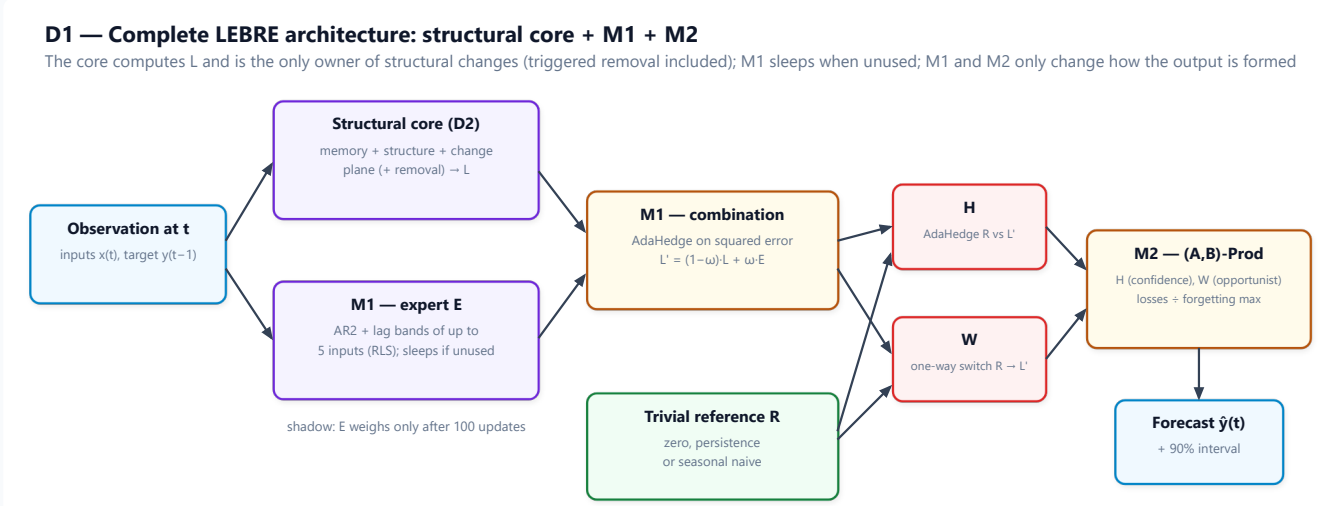
What LEBRE is not

It is not a foundation model nor a long-horizon forecaster; it does not model strongly non-linear relations; it does not claim physical causality (an accepted input is an input with certified *predictive relevance* against the reference, and near-identical inputs are reported as a group). The name comes from the project's original family of architectures (*Lifecycle-governed Evidence-Based Resource Evolution*): structure that enters and leaves the model by evidence, within a budget.

3. Design principles

Principle	Meaning
Every structural change is an experiment	Structure only changes when an anytime-valid sequential test rejects “the change does not improve the forecast”; each acceptance has an error level fixed beforehand.
Unit = a whole input	The question is “does this input help?”, not “does this lag help?”: fewer hypotheses, more power, the response being refined later by hierarchical splits.
Start from the trivial	The final forecast starts from a declared trivial reference and only comes to trust the adaptive part when it proves better on the data themselves.
Layers that do not touch the core	M1 and M2 only change how the final forecast is formed; the structural decisions are the same with or without them (verified on every evaluated series).
Spend only where there is use	A part that is not contributing comes to cost little, but keeps measuring whether it became useful again: M1 sleeps while its weight is negligible and wakes by evidence; the equivalence report uses one sample every 31 steps.
Structure also leaves	When a unit's contribution drops, a removal test with its own evidence (only from the steps of the episode) decides whether it is still supported; removing requires at least as much evidence time as an addition may use.
Fixed, declared cost	Fixed per-step cost given the number of inputs, counted operation by operation, with limits declared before each evaluation; the result is reported even when the limit is not met (§12).
No per-series tuning	The same constants on any series; the environment only provides the period(s) and the trivial reference.
Silence and out-of-contract data are not evidence	Without an observed target, with a clipped input ($|x| > 8\sigma$) or in quarantine, nothing in the structure learns and no evidence accrues.
Output within contract	The core forecast never leaves an envelope of already observed values, and target gaps do not close the memory loop.
Observability	Every change is logged with time, kind, evidence and group of equivalent inputs; the weights of M1 and M2 are readable at every step; the forecast comes with an adaptive interval.

4. Architecture overview

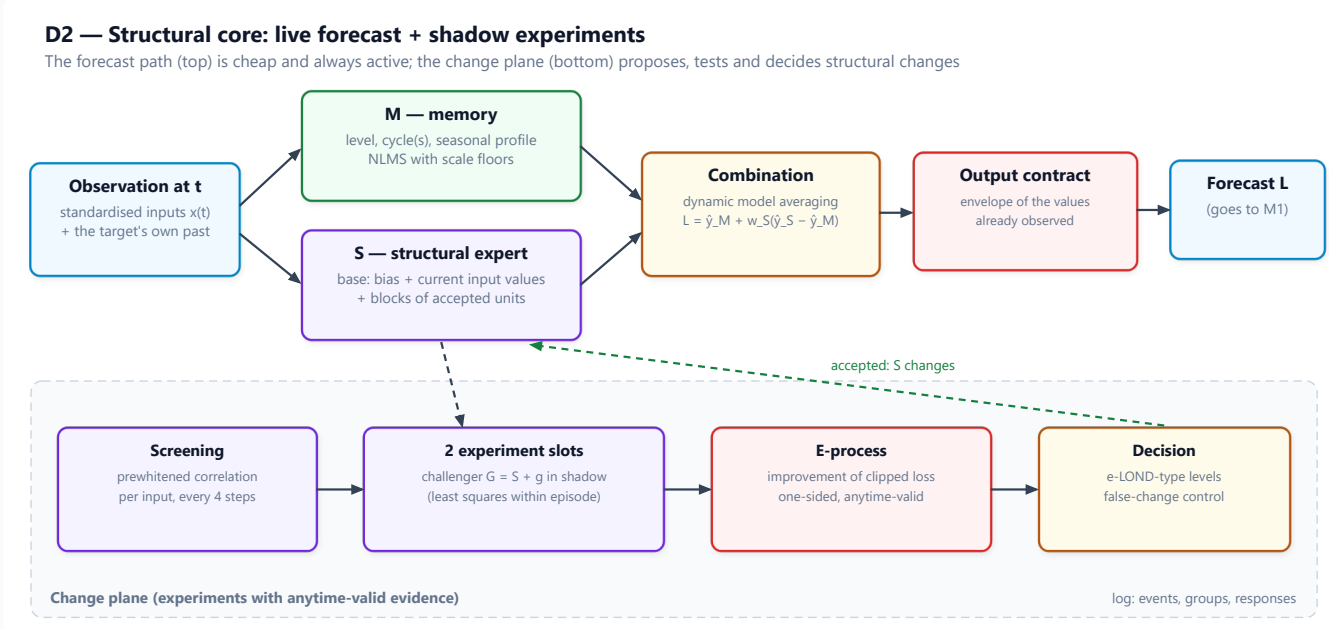


Complete view of one forecasting step. The core runs entirely underneath; M1 forms L' from L; M2 forms the output from L' and the reference R.

LEBRE has three layers, executed in this order at every step:

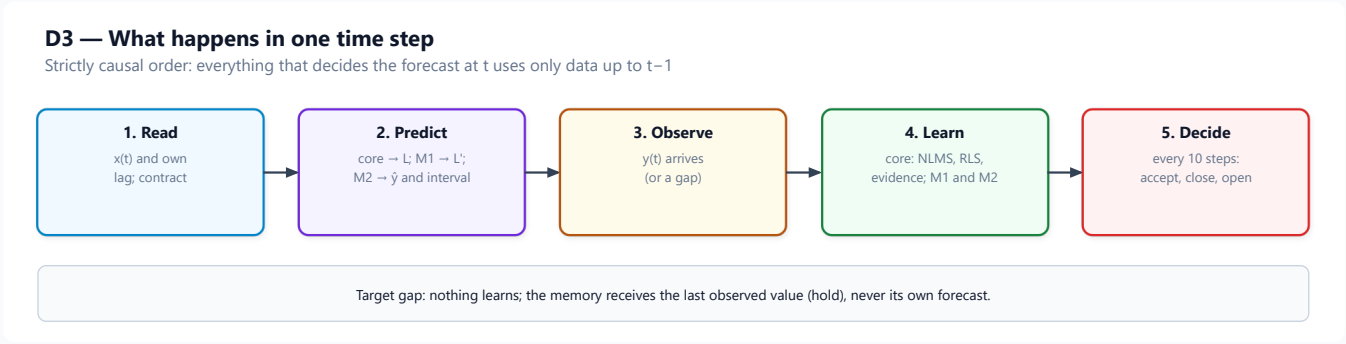
1. **Structural core** (D2): produces the forecast L and is the only owner of structural changes. It has two experts and a combination on the forecast path, and a change plane in shadow.
2. **M1 — precision expert** (§9): a recursive regression E on the target's past and on lag-band means of the most correlated inputs, combined with L by AdaHedge on the squared error: L' = (1 − ω)·L + ω·E. Where E does not help, ω goes to zero and L' = L.
3. **M2 — starting from the trivial reference** (§10): mixes R and L' through two paths — H, conservative (AdaHedge), and W, which switches from R to L' once (switch distribution) — and combines both with an (A,B)-Prod, which guarantees not falling far behind H.

Three mechanisms act inside those layers and do not change what LEBRE predicts where the part concerned is being useful: **M1 dormancy** (§9), the **triggered removal** in the core's change plane (§5.9, §7) and the **cheap equivalence report** in the core (§12, §13). While M1 does not sleep and no triggered removal opens, the forecasts are identical to those of the architecture without these mechanisms.



Structural core. The two experts and the combination form the path of the forecast L. The change plane runs in shadow and only alters S when a change is accepted.

Memory (M). Predicts the target from its own past, relative to the last value: reversion to a slow mean, the last difference, seasonal terms of the declared cycle(s) and a profile of increments per cycle phase. **Structural expert (S).** Linear regression on the current value of each input (the "base") plus the *response blocks* of the accepted units. **Combination.** Dynamic model averaging between M and S, with a weight that follows which of the two has been erring less. **Change plane.** A cheap screen points to the most promising candidate; up to two experiments run in shadow; an e-process accrues the evidence of each change; an online level procedure decides.



Order of operations in one step. The forecast at t is issued before y(t) is read; M1 and M2 learn after the core, with the same y(t).

5. Theoretical basis of the structural core

This section gathers the theoretical basis of the core. **All tools come from the literature** (cited in each item); what is specific to LEBRE is the way they are combined and the verified conditions of use (§8).

5.1 Online linear learning: NLMS and recursive least squares

The live model (base of S, block weights and memory) learns by **NLMS** (normalised least mean squares; Widrow & Stearns 1985; Haykin 2002):

$$w_{t+1} = w_t + \mu \, \frac{e_t}{\|\phi_t\|^2 + \delta} \, \phi_t, \qquad e_t = y_t - w_t^\top \phi_t.$$

In steady state, the **misadjustment** (relative excess squared error caused by the weight jitter) is approximately $M \approx \mu/(2-\mu)$. The core uses μ = 0.05 in S and in the memory.

The **challengers** use recursive least squares (RLS) without forgetting within each episode, with $P_0 = 10\,I$: with few parameters (5 to 7), RLS converges in tens of samples and gives an almost constant coefficient within the episode, which matters for the condition of use (§8). M1's precision expert also uses RLS, with forgetting (§9).

5.2 Memory: exponential smoothing in regression form

The memory is a regression of the increment on features of its own past, inspired by the *airline* model (Box & Jenkins 1970) and by the Holt-Winters seasonal state (Hyndman et al. 2008) — an analogy, not an equivalence:

$$\hat{y}_t^M = y_{t-1} + w_1(m_t - y_{t-1}) + w_2(y_{t-1} - y_{t-2}) + w_3(y_{t-s} - y_{t-1}) + w_4(y_{t-s+1} - y_{t-s}) + w_5 G_{t \bmod s} + \left[\, w_6(y_{t-s_2} - y_{t-1}) + w_7(y_{t-s_2+1} - y_{t-s_2}) \,\right]$$

m: slow exponential mean (rate 0.01); G: exponential mean (rate 0.1) of the increments observed at the same cycle phase; s_2: second declared cycle (week, for buildings). The NLMS normalisation has scale-relative floors (10^{-3} of the mean feature power and 10^{-4} of the slow scale of y), so that flat stretches do not produce huge steps.

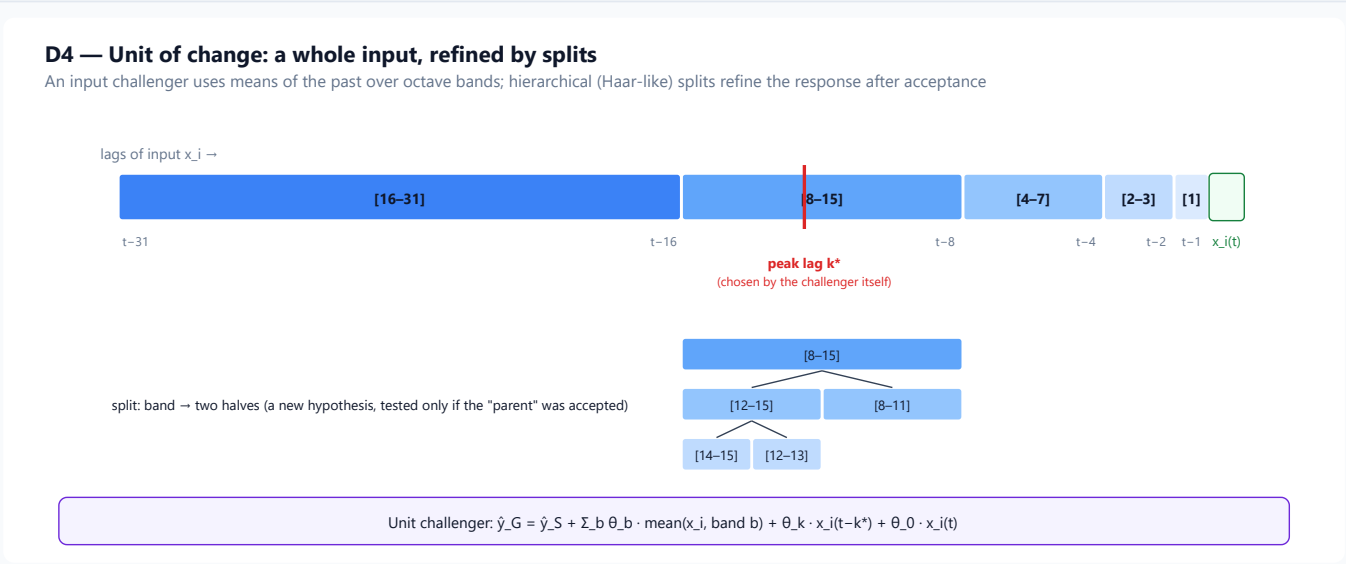
5.3 Combination in the core: dynamic model averaging

The weight of S follows the discounted loss difference between the two experts (Raftery, Kárný & Ettler 2010; a logistic form equivalent to exponential expert weights):

$$D_t = 0.99\, D_{t-1} + \frac{(y_t - \hat{y}_t^S)^2 - (y_t - \hat{y}_t^M)^2}{\hat{\sigma}^2}, \qquad w_S = \frac{1}{1 + e^{\,0.5\, D_t}}, \qquad L_t = \hat{y}_t^M + w_S(\hat{y}_t^S - \hat{y}_t^M).$$

5.4 Response representation: octave bands, peak lag and Haar splits

The response of an input is approximated by means of its past over dyadic bands [1], [2–3], [4–7], [8–15], [16–31] — a resolution that decreases with delay, as in multiresolution decompositions (Mallat 1989) and distributed-lag models. A band mean dilutes an isolated delay (it keeps ~1/width of the effect); therefore the challenger also carries its **peak lag** k*, chosen by itself as the largest prewhitened correlation in the first 200 samples of the warm-up — before any evidence, which keeps the challenger predictable. After acceptance, each band may be **split** into its two halves (one Haar coefficient); each split is a new hypothesis, tested only if the "parent" was accepted, in the spirit of hierarchical testing (Meinshausen 2008; Yekutieli 2008).



Representation of the response to an input: means over octave bands, the current value and a peak lag; after acceptance, each band may be split into halves, one hypothesis per split.

5.5 Screening: prewhitening and group statistic

To choose the next candidate without testing them all, each inactive input has a group statistic: exponential correlations (weight 0.02) between the residual of S and the band means, both **prewhitened** by a causal AR(1) of the input (Box & Jenkins 1970), squared, summed and normalised by the null variance. The unit is opened if the statistic exceeds the 99% quantile of χ^2_5 (15.09); a split, the 99% quantile of χ^2_1 (6.63), on the Haar detail. The screen **produces no evidence**: it only chooses what to test, and the error control comes entirely from the e-process.

5.6 Anytime-valid evidence: e-processes of predictive improvement

An **e-process** is a non-negative process whose expectation under the null does not exceed 1 at any stopping time; by Ville's inequality, the probability that it ever exceeds 1/α is ≤ α. This allows looking at the evidence at every step, pausing and resuming without inflating the error (Ramdas, Grünwald, Vovk & Shafer 2023; Grünwald, de Heide & Koolen 2024). The core uses the sequential comparison of forecasters of Choe & Ramdas (2024), with the *weak* null: on average, the challenger does not improve the loss of the reference by more than a margin ε.

Clipped loss and differential, with the scale $B_t = 2\max(\hat{\sigma}_t^S,\ 0.1\, s_t^{lenta})$:

$$\ell_t(u) = \min\{(y_t - u)^2 / B_t^2,\ 1\}, \qquad d_t = \ell_t(\hat{y}_t^S) - \ell_t(g_t) - \varepsilon.$$

Sub-exponential mixture e-process, with predictable centring $\gamma_i = \min(\bar{d}_{i-1}, 0)$ and 8 values of λ:

$$E_n = \frac{1}{8}\sum_{j=0}^{7} \exp\left\{\lambda_j \sum_{i\le n} d_i - \psi_{E,c}(\lambda_j) \sum_{i\le n} (d_i - \gamma_i)^2\right\}, \quad \psi_{E,c}(\lambda) = \frac{-\log(1 - c\lambda) - c\lambda}{c^2}, \quad \lambda_j = \frac{0.9 \cdot 2^{-j}}{c}.$$

The one-sided constant c. The statement of Choe & Ramdas assumes |d_i − γ_i| ≤ c, which would give c = 2(1+|ε|). The underlying proof (Howard et al. 2021, §A.8) only uses the lemma of Fan, Grama & Liu (2015): exp{λξ − ψ_E(λ)ξ²} ≤ 1 + λξ for ξ ≥ −1 and λ ∈ [0,1). For the one-sided test, under the weak null E[d_i | past] = μ_i ≤ 0, the **lower** bound d_i − γ_i ≥ −c suffices:

$$\mathbb{E}_{i-1}\left[e^{\lambda(d_i-\gamma_i)-\psi_{E,c}(\lambda)(d_i-\gamma_i)^2}\right] e^{\lambda(\gamma_i-\mu_i)} \le \big(1+\lambda(\mu_i-\gamma_i)\big)e^{\lambda(\gamma_i-\mu_i)} \le 1.$$

Since λΣd_i ≤ λΣ(d_i − μ_i), the process is bounded by a non-negative supermartingale. Since d_i ≥ −(1+ε) and γ_i ≤ 0, $c = 1 + \max(\varepsilon, 0)$. A mixture of e-processes is an e-process, and pauses between episodes form a predictable subsequence.

This observation is a direct reading of the published proof, not a new result. In a development simulation (400 trajectories × 5,000 steps, null at the boundary) the type-I error was ≤ 0.025 against α = 0.05, and power rose from 0.66 (two-sided c) to 0.77 (§15). The betting alternative with predictable λ (Waudby-Smith & Ramdas 2024) had more power but accepted near-identical substitutes of the true input and was not adopted.

5.7 Multiplicity over the stream: e-LOND-type levels

Each change hypothesis receives, at creation, a fixed level that depends only on the number of hypotheses already created and of changes already accepted (the LOND rule of Javanmard & Montanari 2018, in the e-value version of Xu & Ramdas 2024). This controls the false change rate (FDR) over an endless stream, with asynchronous decisions:

$$\alpha_n = \min\big(\alpha\,\gamma_n\,(R_n+1),\ 0.5\big), \qquad \gamma_n = \begin{cases} 1/(2P) & n \le P \\ \frac{1}{2}\,C/\big((n-P)\log^2(n-P+1)\big) & n > P \end{cases}, \qquad \text{accept if } \log E \ge \log(1/\alpha_n),$$

with α = 0.05, P = 2 × (number of units) and C such that Σγ_n ≤ 1. A hypothesis is *persistent*: its evidence accumulates across episodes.

5.8 Adaptive interval

The 90% interval is a quantile of the absolute error updated by coverage feedback, in the spirit of adaptive conformal inference (Gibbs & Candès 2021): every 8 steps, $\hat{q} \leftarrow \max(0,\ \hat{q} + 0.1\cdot\hat{\sigma}\Sigma(\text{miss} - 0.1))$. The same rule is applied to the error of the final output (§11). There is no formal coverage guarantee under dependence; empirical coverage is reported.

5.9 Removal with per-episode evidence

The removal hypothesis of a unit is persistent (§5.7): it accumulates the evidence of all episodes since the acceptance. In a series with a regime change, the evidence from before the change, when removing was clearly worse, weighs forever, and removal never reaches the threshold. The null "on average over the whole tested past" is the wrong null for removing. Therefore, besides the periodic schedule (which stays the same), the change plane opens a **triggered removal test** when the unit's mean contribution drops below half of its reference, with a new hypothesis per episode: "over the steps of this episode, removing does not improve the clipped loss by more than the margin".

- **Validity.** The opening rule uses only the past; the e-process starts from zero in each episode and is the same as in §5.6. Accepting only after a minimum number of evidence steps is a valid stopping time: Ville's inequality holds at any stopping time.
- **Own budget.** Each triggered test receives a level α·γ_k·(R^rem + 1), with α = 0.05, γ_k = C/(k·log²(k+1)) and R^rem the triggered removals already accepted; k counts only those tests. The add, swap and split hypotheses keep the budget of §5.7, without counting removals: the guarantee on false removals (removing what worsens more than the margin) is separate from the one on false additions.
- **Why it is slow.** After a change, the per-step gain of removing is small on the scale of the clipped loss (~0.002 beyond the margin), and the optimal λ lies above the e-process grid; evidence grows by ~0.001 per step. That is why the removal episode may last up to 10,000 steps, and acceptance requires at least 5,000 (the maximum time of an addition episode).

6. Theoretical basis of M1 and M2: aggregation of forecasters

M1 and M2 are problems of **prediction with expert advice**: at every step two forecasts are available and one must decide how much weight to give each, using only the past. Three algorithms from the literature are used, without modifying their cores; LEBRE's choices are which losses they receive and how they are chained.

6.1 AdaHedge

Exponential weights on the cumulative losses with a learning rate tuned by the data themselves (de Rooij et al. 2014): it is the logarithm of the number of experts divided by the cumulative sum of the *mixability gaps* (how much the loss of the mixture exceeded the ideal "mix" loss). The regret against the best expert grows as the square root of the loss variation, and the algorithm is **invariant to translation and scale** of the losses: it does not need to know the error scale, which allows using the squared error without clipping or normalisation.

$$w_{k,t} = \frac{e^{-\eta_t S_{k,t-1}}}{\sum_j e^{-\eta_t S_{j,t-1}}}, \qquad \eta_t = \frac{\ln K}{\Delta_{t-1}}, \qquad \Delta_t = \Delta_{t-1} + \left(\sum_k w_{k,t}\ell_{k,t} + \tfrac{1}{\eta_t} \ln \sum_k w_{k,t} e^{-\eta_t \ell_{k,t}} \right),$$

with S_k the cumulative loss of expert k and K = 2. When Δ = 0, the weights follow the leader.

6.2 Switch distribution with one switch

The *switch distribution* (van Erven, Grünwald & de Rooij 2012) is a Bayesian mixture over expert sequences that allows switching expert over time. LEBRE uses the simplest version: three hypotheses — "always R", "always L'" and "R until some time, L' after" —, with equal priors and switch hazard 1/(n+1) at time n. The posterior is updated by the Gaussian likelihoods of the two forecasts, with a common variance equal to the smaller recent squared error of the two (with a floor). It concentrates weight quickly on the reference when the adaptive part does not help, and switches once when the adaptive part becomes clearly better. When all the mass is on one side, the direct update can give 0/0 by *underflow*; in that case the same computation is done in log scale.

6.3 (A,B)-Prod: not falling far behind a trusted expert

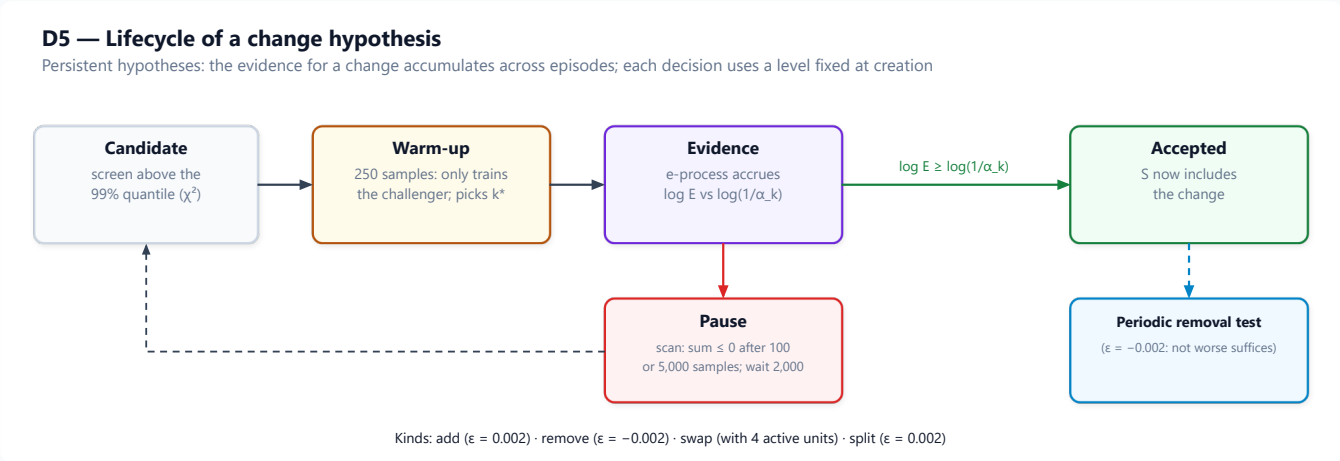
(A,B)-Prod (Sani, Neu & Lazaric 2014) combines an *opportunistic* algorithm A with a *trusted* algorithm B. B's weight is fixed at 1/2 and A's is updated multiplicatively by the loss difference; the fraction given to A is s_t. For losses in [0, 1], it guarantees **constant** regret against B and of order √(T log log T) against A, in the version without a known horizon. This kind of asymmetric guarantee — almost nothing to lose against the safe choice — comes from the Pareto frontier of regret (Even-Dar et al. 2008; Koolen 2013).

$$s_t = \frac{\eta_t w_t^A}{\eta_t w_t^A + w^B/2}, \qquad w_{t+1}^A = w_t^A \, (1+\eta_t \delta_t)^{\eta_{t+1}/\eta_t}, \qquad \delta_t = \ell_t^B - \ell_t^A, \qquad \eta_t = \min\left(\tfrac{1}{2}, \sqrt{1/(1+\textstyle\sum_{\tau<t}\delta_\tau^2)}\right).$$

6.4 The limit of unbounded losses

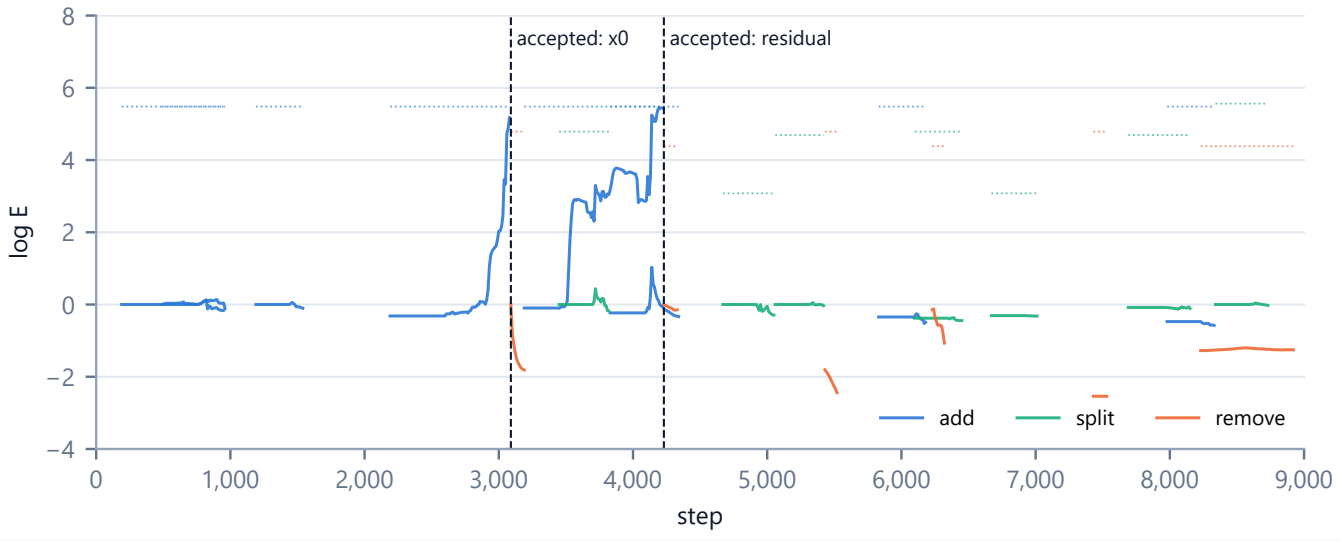
The squared error has no upper bound. The known guarantees for unbounded and heavy-tailed losses (Cesa-Bianchi, Mansour & Stoltz 2007; Hazan & Kale 2010; Orabona & Pál 2018; Mhammedi, Koolen & van Erven 2019; V'yugin & Trunov 2019; Alquier 2021; Moulin, Esposito & van der Hoeven 2025) hold in units of the **largest single-step loss**. When a single step accounts for a large part of a series' whole error — as in an FX shock —, no online combiner can guarantee tracking the best expert with a small slack relative to the total error. This limit is a property of the data, not of the algorithm, and is handled explicitly in the evaluation (§10, §14).

7. Lifecycle of a structural change

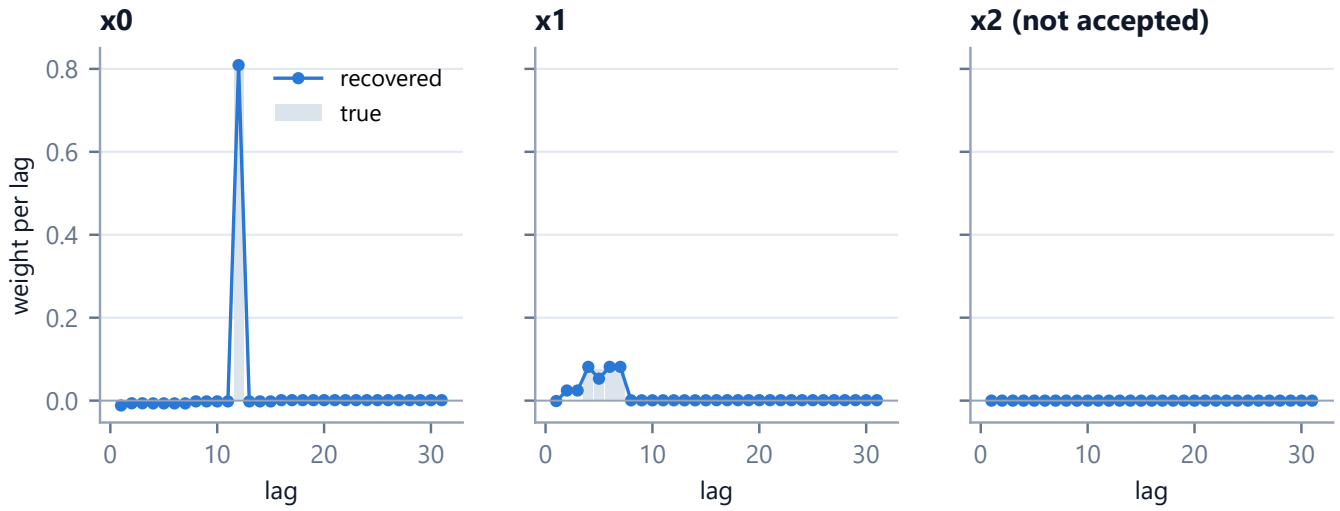


States of a change hypothesis. Evidence only accrues outside the warm-up, with an observed target and outside quarantine.

- **Units:** each input (including the target's own past, standardised causally) and the residual latent state (two filters of the standardised error, poles 0.8 and 0.95). Up to 4 active units.
- **Input challenger:** RLS on [means of the 5 bands, peak lag, current value x_i(t)] — the block is judged net of what x_i(t) already carries. The RLS target is the residual the change would explain.
- **Episode:** 250-sample warm-up without evidence; ends on acceptance, by the scan rule (episode sum ≤ 0 after 100 samples) or at 5,000 samples; the same hypothesis waits 2,000 steps before returning. Only one challenger warms up at a time (cost-peak control).
- **Decisions** every 10 steps: accept, close, open. Priority: triggered removal test (§5.9), when the contribution of an active unit has dropped below 0.5 times its reference (the peak since the acceptance, which resets to the current value at each triggered test), with at least 2,000 steps between triggered tests of the same unit — it closes a periodic removal test of the same unit, whose persistent hypothesis is kept; then the periodic removal test; then the best screened candidate (unit or split). With 4 active units, the candidate enters by *swap* with the one of smallest contribution.
- **Judged on the forecast of S**, not on the combination nor on the final output: the question is whether the change improves the structural expert. M1 and M2 take no part in structural decisions.



Real example (development series: flow at a hydro plant; input x0 = outflow of the upstream plant; x1 = the target's own past). Log-evidence of the experiments (solid) against their thresholds log(1/α_k) (dotted). Input x0 is accepted at t ≈ 3,090 and the residual latent state at t ≈ 4,230; afterwards, removals and splits are tested and closed without evidence.



Declared synthetic process (y = 0.8·x0(t−12) + 0.3·mean(x1, t−4…t−7) + noise; x2 irrelevant). Response recovered by the core (dots) against the true one (bars): lag 12 of x0 is recovered with weight 0.81; the distributed effect of x1 appears in band [4–7]; x2 is not accepted.

8. The condition of use: reference misadjustment

A test of predictive improvement compares **methods** (model + estimation procedure), not models — a distinction known in econometrics (Giacomini & White 2006). During development, this appeared concretely: with the base of S learning by NLMS with step 0.1, targets **without any structure** had a change accepted in ~20% of the runs. The cause is that the estimation error of the reference is predictable from the past, and a challenger can "correct" it.

Let $f_t^* = \mathbb{E}[y_t \mid \mathcal{F}_{t-1}]$, $\eta_t = f_t^* - S_t$ (estimation error of the reference, predictable) and the challenger $G_t = S_t + g_t$. For the normalised squared-loss difference $\Delta_t = [(y_t - S_t)^2 - (y_t - G_t)^2]/B_t^2$:

$$\textbf{Proposition 1} \quad \mathbb{E}[\Delta_t \mid \mathcal{F}_{t-1}] = \frac{2g_t\eta_t - g_t^2}{B_t^2} \;\le\; \frac{\eta_t^2}{B_t^2}, \quad \text{with equality at } g_t = \eta_t.$$

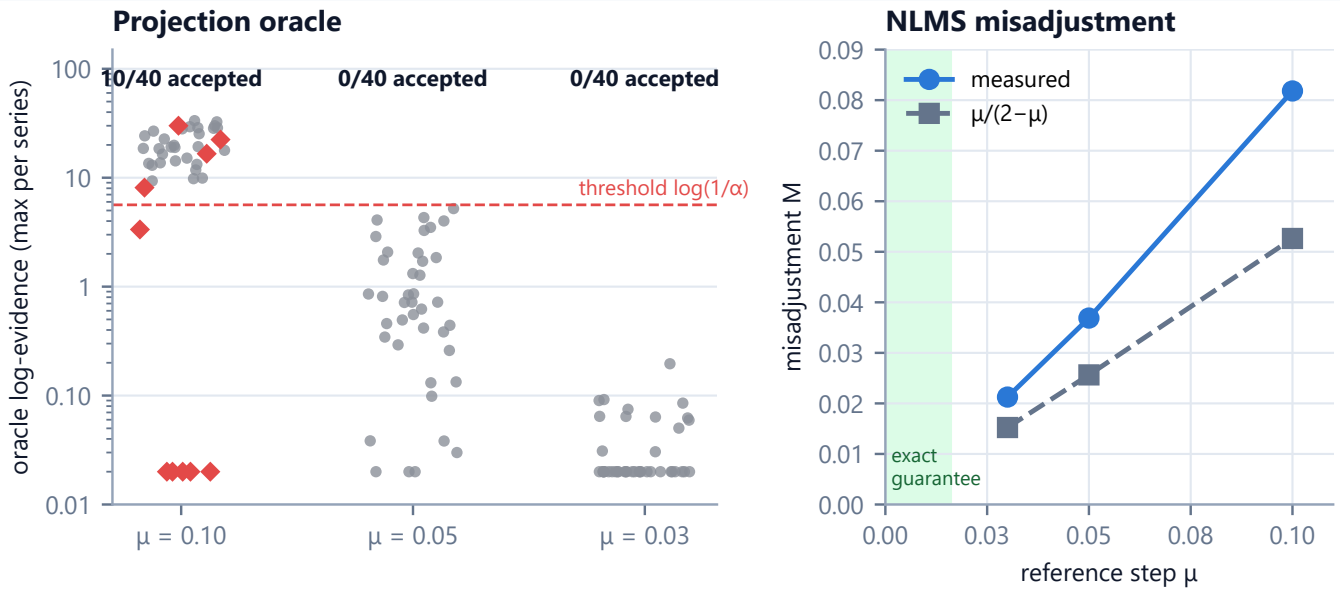
Proof: y − S = ε + η and y − G = ε + η − g; expanding, the difference is 2g(ε+η) − g², and E[ε | past] = 0. ■

$$\textbf{Corollary (exact guarantee)} \quad \frac{1}{n}\sum_{i\le n}\frac{\eta_i^2}{B_i^2} \le \varepsilon \;\; \forall n \;\Rightarrow\; \text{under the structural null, the test null holds.}$$

For NLMS with misadjustment M, this requires $M \lesssim 4\varepsilon$ (with ε = 0.002: μ ≲ 0.016).

$$\textbf{Proposition 2 (fixed coefficient)} \quad \text{mean improvement over the window} \le D_u + \varepsilon, \quad D_u = \tfrac{1}{|W|}\textstyle\sum_{i\in W}(P_u\eta_i)^2/B_i^2 - \varepsilon,$$

where P_u is the projection of η on the unit's features. **Proposition 3 (practical criterion, approximation):** log-evidence grows as n·D²/(2σ_d²); on a series of length T, false acceptances are unlikely if $D_u < \sigma_d\sqrt{2\log(1/\alpha_k)/T}$.



Development check (120 runs with structureless targets, real inputs). Left: log-evidence of the best possible challenger (projection oracle) against the threshold; actual acceptances in red. With μ = 0.1 the oracle crosses the threshold (median 18.2) and there were 10/40 acceptances; with μ = 0.05 (max 5.2) and μ = 0.03, none. In runs that accepted, the oracle was measured after the acceptance and may be low (points at the floor). Right: measured and theoretical misadjustment.

Honest reading: the core uses μ = 0.05, which lies in the **practical regime** of Proposition 3 (the deviation from the null exists, ~0.0013, but is smaller than what is detectable within the series length, ~0.0019), and **not** in the exact-guarantee regime (μ ≲ 0.016). Therefore false-change control is checked empirically with null batches (0/40 and 0/30 acceptances in development), not only asserted by theory. The observation relates to Giacomini & White (2006) and, through another mechanism, to Han & Qu (2026); we did not find it stated explicitly for this situation, but it is a simple consequence of known results.

9. M1: precision expert and dormancy

Motivation. The core estimates the target dynamics (memory) and the input effect (structure) separately, with cheap learning. In some families this leaves it **below the accuracy × cost frontier**: a regression of equal or lower cost, estimating both jointly, errs 5% less (§17). M1 adds that joint estimation as a second expert, without touching the core.

The expert E

Variables: the target itself at lags 1 and 2 and, for each of the m = min(d, 5) inputs with the largest absolute correlation with the target (d = number of inputs), the means over the lag bands [0], [1], [2–3], [4–7] and [8–15]:

$$z_t = \left[1,\ \tilde{y}_{t-1},\ \tilde{y}_{t-2},\ \bar{x}_{i,b}(t)\right]_{i\le m,\, b\le 5}, \qquad k = 3 + 5m, \qquad E_t = \mathrm{clip}\left(w_t^\top z_t,\ L_t \pm 2\max(\hat{\sigma}_L, \mathrm{floor})\right)$$

- **Input selection:** exponential means (rate 0.01, past only) of x, x^2, x·y, y and y^2, updated every 8 learned targets; the selection is made after 100 targets and redone every 5,000. The means and deviations at selection time standardise the chosen inputs and the target. If the set changes, the regression and the combination restart.
- **Missing inputs** are replaced by the last value seen (0 before the first).
- **Weights:** recursive least squares with forgetting λ = 0.999 and $P_0 = 100 \cdot I$, updated every 8 learned targets (outside quarantine and with inputs within the contract); the matrix is symmetrised at every update and reset if it loses positive definiteness. E only exists after k updates (determined regression).
- **Clip:** E stays within L ± 2·max($\hat{\sigma}_L$, floor), with $\hat{\sigma}_L$ the recent root mean squared error of the core (exponential mean, rate 0.01) and the floor 0.1 times the slow scale of the target; this bounds the damage of an ill-conditioned regression.

The combination with the core

AdaHedge (§6.1) with two experts, L and E, on the **squared error without clipping or normalisation**: the weights follow the accumulated squared error, which is what the mean squared error measures, and the learning rate adapts to the scale by itself. The combination observes every observed target at which E exists. **Shadow entry:** until 100 updates, L' = L.

$$L'_t = (1-\omega_t)\,L_t + \omega_t\,E_t, \qquad \omega_t = \frac{e^{-\eta_t S^E_{t-1}}}{e^{-\eta_t S^L_{t-1}} + e^{-\eta_t S^E_{t-1}}}, \qquad S^{\bullet}_{t-1} = \sum_{\tau<t}(y_\tau - \bullet_\tau)^2$$

Why AdaHedge and not an (A,B)-Prod: Prod needs losses in [0, 1]. Three ways of putting the squared error into that interval were measured and failed for different reasons (§17): clipping at 2σ leaves Prod blind to large errors; dividing by the largest error ever seen freezes it after a record; dividing by the largest recent error gives small weight to the large-error steps, exactly where the expert gains. Trading Prod's constant guarantee for AdaHedge's (a regret that, relative to the total error, vanishes with series length) is what worked.

Dormancy: cost only where there is use

In many real series, M1's weight ω goes to zero: M1 is paid at every step without changing the output. In the previous version's final evaluation this happened on 70 of 104 series. M1 therefore has two modes.

- **Active:** exactly as described above.
- **Dormant:** entered after the shadow, when $\omega < 10^{-3}$ for 2,000 consecutive learned targets. The output is L' = L and AdaHedge is frozen. The forecast E is computed only every 8 targets, to measure the relative gain $g = ((y-L)^2 - (y-E)^2)/(y-L)^2$, and the regression is updated only every 64 targets, so that E does not go stale.
- **Waking:** when the exponential mean (rate 0.01) of g exceeds 0.05 after at least 100 samples. AdaHedge restarts with a new 100-update shadow, like a new expert.
- **Short trial on reselection:** if M1's input selection changes during sleep, the expert is new; M1 wakes in trial and goes back to sleep after 500 consecutive targets with $\omega < 10^{-3}$, unless ω reaches 10^{-3}.

The rule uses only the past. While M1 does not sleep, the output is identical to M1's without dormancy. Sleeping is a bet that M1 would remain weightless; the cost of losing that bet is the delay until waking, measured in development and in the audit (§17, §19.5).

10. M2: starting from the trivial reference

Motivation. In nearly unpredictable series (for example, FX returns and levels), an adaptive part may err more than predicting zero or repeating the last value: it learns noise. M2 makes the output start from a **trivial reference R** declared with the data — zero for changes such as financial returns, seasonal naive (the value one cycle ago) when there is a cycle, persistence (the last value) otherwise — and only come to trust L' when it proves better. While R is undefined (before a full cycle, in the seasonal case), the output is L'.

- **H** — AdaHedge between R and L' on the squared error (§6.1): confidence, no switches.
- **W** — one-way switch from R to L' (§6.2), on Gaussian losses $(y-\cdot)^2/2s^2$, with $s^2 = \max(\min(\sigma^2_R, \sigma^2_{L'}), \text{floor}^2)$ and σ^2 exponential means (rate 0.01) of each side's squared errors, past only.
- **Combination** — horizon-free (A,B)-Prod (§6.3) with B = H (confidence) and A = W (opportunist); the losses are $(y-\cdot)^2/N_t$, with $N_t = \max(\text{step losses}, 0.99 \cdot N_{t-1})$ — the largest recent error, with forgetting, to put the losses into [0, 1] without freezing after a record.
- H, W and Prod only learn on steps with an observed target, outside target quarantine and with R defined.

$$\hat{y}_t = s_t\,W_t + (1-s_t)\,H_t, \qquad H_t = w^H_{R,t}R_t + w^H_{L',t}L'_t, \qquad W_t = w^W_{R,t}R_t + w^W_{L',t}L'_t$$

Gate audit. In parallel, an e-process (§5.6) with its own budget α = 0.01 and margin ε = 0.002 tests whether the adaptive part improves on the current reference (and, after a switch, the opposite); each acceptance is logged as an event and the hypothesis is recreated. This test **does not decide the output**: it only documents when the evidence changed sides, with the same kind of guarantee as the structural changes.

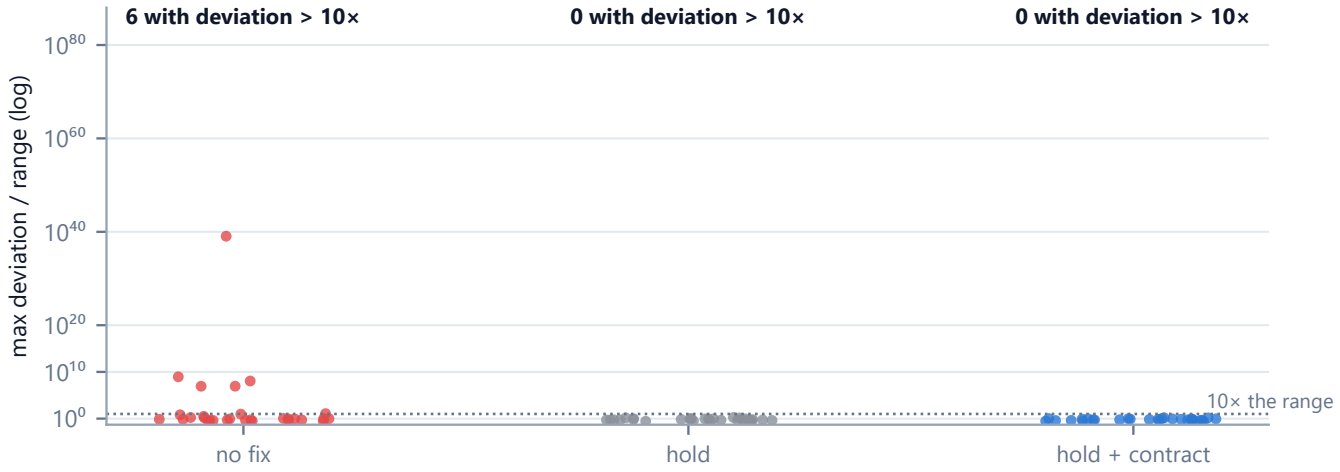
Guarantees and the limit of shocks

For the normalised losses, Prod guarantees constant regret against H and of order $\sqrt{T \log\log T}$ against W; in the original scale there is no clean formal bound, because the normalisation changes over time. The limit of §6.4 shows up in the data: in financial series a single day can account for much of the whole error (in the 521 development series measured, 32 have the largest step above 5% of the reference's total error; the largest, baht_1990, 25.9%). That is why the evaluation ruler treats those series as undecidable (§14).

11. Output safeguards and interval

Two safeguards act on the output path of the core and cost ~15 operations per step. Both are standard engineering practice; they are here because their absence produced the most severe failure observed (§15).

- **Hold during target gaps:** the memory receives the last observed value, never its own forecast. Without this, the memory runs open-loop and, with some learned weights, the recursion is unstable (characteristic root ~1.75 in a real case).
- **Output contract:** the core forecast is limited to the envelope of observed targets (running maximum and minimum relaxing toward the slow mean at rate 10^{-3}), widened by 0.5× its width; a non-finite forecast becomes the last observed value.
- **M1 and M2 inherit the containment:** E stays at most 2·max($\hat{\sigma}_L$, floor) from L; R is an already observed value (or zero); every output of M1 and M2 is a weighted mean of contained forecasts. A non-finite loss never enters the weights.



Stress test of the core in development: four artificial target gaps (24, 100, 400 and 1,500 steps) on 29 series. Maximum deviation of the forecast relative to the observed range. By the criterion declared before the test (deviation > 10×, non-finite forecast or error > 1.5× that of the same series without gaps) there were 10, 2 and 2 series; the two remaining with the fix are a short series (1,024 steps) where the gaps cover almost the whole test region, with deviation < 0.4 of the range.

Output interval

The final output $\hat{y}$ comes with a 90% interval computed by the rule of §5.8 applied to the error of $\hat{y}$ (starting at 1.645 times the recent root mean squared error). The coverage of the core interval was measured in the core's own evaluations (§13); the coverage of the complete output interval **was not reported** in the final evaluations.

12. Computational cost and memory

Cost is counted operation by operation in the code (additions, products, divisions, roots, exponentials and logarithms count 1 FP), per component. The per-step cost is **almost fixed**: memory, base, screening, M1 and M2 always run; the core's experiments add cost only when a slot is busy, and peaks occur at decision steps. Nothing grows with time.

Core

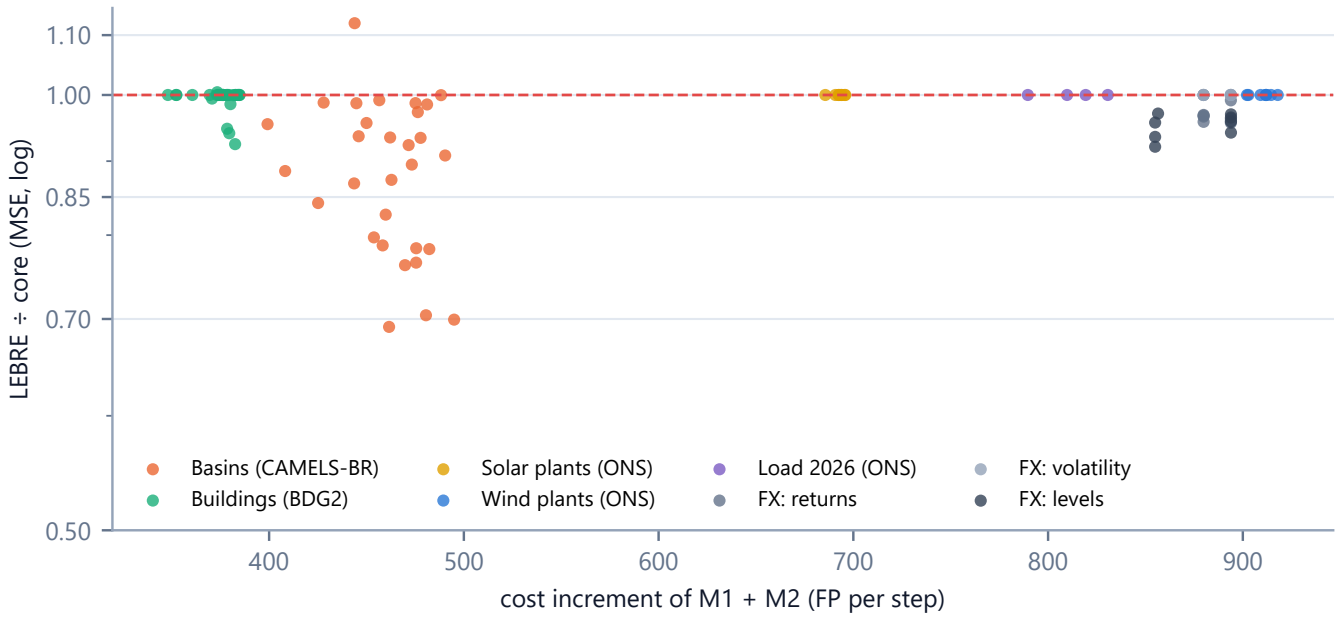
The core was designed with a contract of mean ≤ 400 and maximum ≤ 1,000 FP per step on real series. In its own pre-registered evaluation (60 series), the mean was 415 and the maximum 1,020: the contract **was not met**, by 3.7% in the mean and 2.0% in the peak, and it was not revised after the result. In the previous final evaluation (§19.4), the core cost a median of 461 FP per step.

M1 + M2

The count of M1 and M2 was audited: the implementation performs exactly the declared operations (each pair of the RLS matrix symmetrised once; only the new input row standardised), with forecasts identical to the validated ones in 813 of 813 series; the audit raised the count by 3.2% (median). Per RLS update: $6k^2 + 5k$ (every 8 targets); expert prediction: $2k + 18m + 4$; AdaHedge: 48; M2: ~132 to 137.

Part	FP per step (median, one series per development family, before the audit)
core	453
M2	137
M1: RLS update	513
M1: rest	146
Total	**1,257**

Cost ceiling of M1 + M2. The preliminary design target (total cost ≤ 1.25 times that of the core) proved incompatible with M1 + M2: with the core at ~450 FP, the maximum increment would be ~110 FP, less than M2 alone. Before the previous final evaluation, the project owner replaced it with a ceiling of an **absolute increment of 1,100 FP per step** on every series (the new cost is nearly fixed; a ratio would penalise the series where the core is cheap), plus 2.5 times the core's cost on the families below the frontier. In development (813 series) the increment had a median of 538 and a maximum of 974 FP; in the previous final evaluation, a median of 476 and a maximum of 918 FP, with total cost between 667 and 1,637 FP.



Final evaluation of the previous version (§19.4): cost increment of M1 + M2 and error ratio LEBRE ÷ core, per series. Without dormancy, the cost was paid on every series; the gain appeared in basins and FX.

Memory

The state has a fixed size, given the number of inputs and the periods. The core state was measured in the C port (capacities for up to 8 inputs and 160 hypotheses): 74,988 bytes; most of it is the hypothesis table, which can be reduced. M1 and M2 hold 1,263 numbers (k = 28 in the worst case: 1056 in M1, 207 in M2), 10,104 bytes in double precision. Total **estimated**: 85,092 bytes (83.1 KB), within the pre-registered limit of 128 KB; M1 and M2 were not ported nor measured.

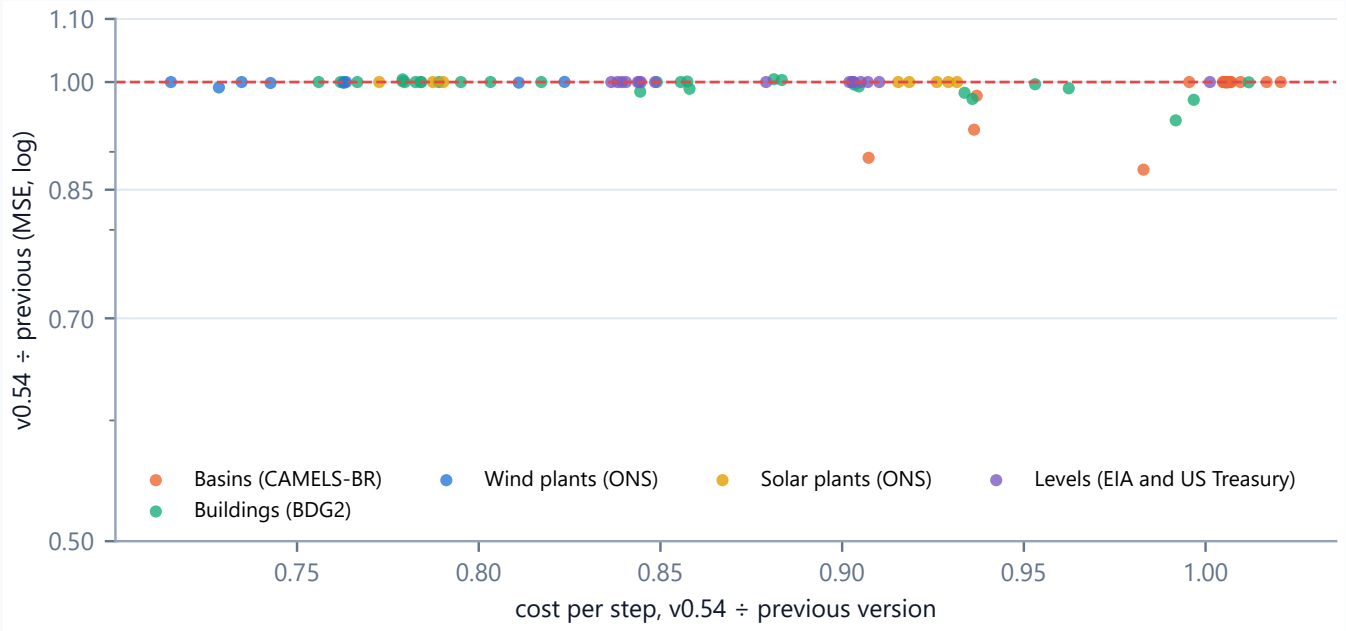
v0.54 adds 22 numbers to the state (M1 sleep and wake; reference and last triggered test per active unit; the sample vector of the equivalence report does not change the order of magnitude): estimated memory 85,268 bytes (83.3 KB).

Cost proportional to use (v0.54)

- **Dormant M1:** while asleep, M1 computes E every 8 targets and updates the regression every 64, instead of predicting at every step and updating every 8.
- **Equivalence report:** the input covariance matrix, used only by the report, had a quadratic cost in d every 4 steps (~1,000 operations per step with 50 inputs). It now uses one sample every 31 steps (a prime, to visit every phase of cycles such as 7, 12, 24 and 168), applied one row at a time over the following steps. The saving is (3d(d+1)/2 + 2d)·(1/4 − 1/31) operations per step; with 50 inputs, the core cost dropped to 0.719 times the previous one in validation 7.
- **Triggered removal:** 2 operations per active unit at each decision, plus the test when it opens.

Family	series	cost per step, previous version (median)	v0.54 (median)	per-series ratio (median)	series where M1 slept
Basins (CAMELS-BR)	30	863	867	1.006	4
Buildings (BDG2)	30	809	675	0.847	28
Wind plants (ONS)	8	1,521	1,187	0.753	8
Solar plants (ONS)	8	1,207	1,081	0.917	8
Levels (EIA and US Treasury)	17	1,105	966	0.879	16

In the final evaluation, the median per-series cost ratio was 0.905 [0.857; 0.932]. Where M1 is useful (basins), the cost is the same as in the previous version; where it is not, it drops. In the audit, wall-clock time followed the count (median of 0.845 times that of the previous version).



Final evaluation of v0.54: cost ratio and error ratio v0.54 ÷ previous version, per series.

Per-step peak (errata 01). In the promoted version, the whole update of the report matrix, every 31 steps, periodically coincided with M1's regression and raised the per-step peak: up to 1.522 times that of the previous version with 50 inputs. The fix spreads the same update over rows; forecasts, decisions, report and total cost stay identical, and the peak came to at most 1.054 times (§19.5).

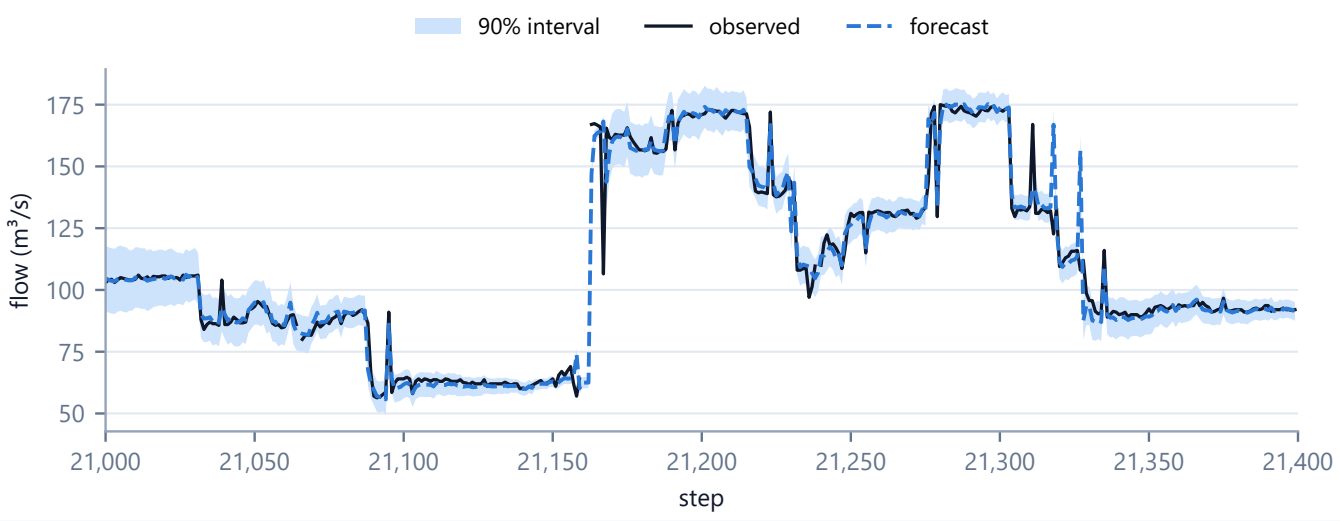
13. Observability and explainability

Each forecast can be decomposed into named contributions: in the core, memory, base and each active block; then, M1's weight ω and, in M2, the weights of R and L' in H and in W and the fraction s_t. Every structural change and every acceptance of the gate audit is logged. Example from a core development series:

Time	Event	Change	log E	Evidence samples
3,090	accepted	add input x0	7.18	957
4,230	accepted	add residual latent state	5.49	893

- **Equivalent groups:** together with the accepted input x0 the model reports the group {x0, x1} (causal correlation ≥ 0.95 between x0, the outflow of the upstream plant, and x1, the target's own past). The claim is "this input or an equivalent one".
- **90% interval of the core:** empirical coverage of 0.906 on this series and 0.904 on average over the 60 series of the core's own evaluation.
- **Log of closed experiments:** each test that did not pass is logged with its evidence (for instance, removal tests of x0 with log E between −1.8 and −3.2: the input remains useful).
- **M1 and M2:** M1's weight ω is read at every step, together with its state (active, asleep, in trial; how many times it slept and woke; M1 cost per step); which inputs M1 chose is also available.
- **Removals:** each triggered removal test is logged like the other experiments, and a periodic test it closes appears as superseded.

- **Equivalence report:** the query first applies the pending rows of the last sample; the report is the same at any time. With one sample every 31 steps, the estimated correlation near the 0.95 threshold is noisier (standard error ~0.017 vs ~0.006); away from the threshold nothing changes (§17).



A 400-step window of the same series: observed value, core forecast and 90% interval.

14. Evaluation methodology

Two questions were evaluated with their own protocols, both pre-registered: (a) is the core competitive within its budget class? (b) do M1 and M2 improve the core without harming any domain?

Common discipline

- **Online one-step forecasting**, without looking ahead; all models receive the same inputs, standardised causally.
- **Pre-registration:** before each access to reserved data, a document fixed questions, metrics, comparators, criteria and the code hashes; each reserve was run a single time; the data loader blocks access to the reserves outside a pre-registered run. Infrastructure incidents (without any model change) are described with the results.
- **Series split** by fixed-seed permutations, before any modelling; each validation set was chosen from metadata only, frozen with SHA-256 and used once.

(a) The core in the budget class

- **Metric:** mean squared error on the test region (last 70% of each series, observed target), divided by that of the calibrated online NLinear on the same series; geometric mean across series; 95% CI by bootstrap stratified by domain (2,000 resamples). **Catastrophic failure:** relative error > 10, deviation > 10× the observed range, or a non-finite forecast.
- **Data:** flows and generation from the Brazilian National Grid Operator (ONS; cascaded plants), CAMELS-BR basins (Chagas et al. 2020; precipitation, evapotranspiration, temperature) and Building Data Genome 2 meters (Miller et al. 2020; weather), plus two system-identification benchmarks in development.
- **Foundation models** (too expensive to run at every step): evaluated on 1,000 points per series, the same for all, with the same context and the current inputs as known future covariates. **Batch-fitted comparators** (SARIMAX with inputs, FITS, SparseTSF, TTM fine-tuning): fitted on the first 15% of the series (up to 5,000 points), then with fixed weights.

Set	Series	Use
Development	29 (ONS 11, CAMELS 10, BDG2 5, Silverbox, Cascaded Tanks ×2)	all design and tuning of the core
Reserve 1	125 (ONS 45, CAMELS 50, BDG2 30)	revealed the gap failure (§15)
Reserve 2	40 (CAMELS 20, BDG2 20)	checked the fix
Reserve 3	60 (CAMELS 30, BDG2 30)	ultralight comparators, TTM, C port, microcontroller

(b) M1 and M2 against the core

The final-evaluation data were reserved on 2026-10-05, before any M1 and M2 code, with hashes and locks. The metric is the mean-squared-error ratio LEBRE ÷ core, series by series. **Decidability ruler:** each series is classified by the 95% CI of core ÷ R (paired block bootstrap, 2,000 replicates) as "core better", "reference better" or "undecidable". Criteria: (1) on "reference better", LEBRE ÷ R with upper limit ≤ 1.01; (2) on "core better", LEBRE ÷ core with upper limit ≤ 1.02; (4) on negative tests, no more false inputs than the core; (6) the core's structure preserved; (7) on short series (360 steps), ≤ 1.05 of the best of core and R. **Shock addenda** (decided during development, before validation 5 and the previous final evaluation): a series whose largest single step exceeds max(5%, 5·2 ln(n)/n) of R's error is undecidable on criteria 1 and 2 (n = evaluated steps), because on it no combiner has a useful guarantee (§6.4).

F6, on the families where the core lies below the accuracy × cost frontier: cost ≤ 2.5 times that of the core and error ≤ 1.05 times the lowest error among the diagnostic's 8 regressions with cost ≤ that of LEBRE. **Promotion** (pre-registration; project-owner decisions): (A) no harm per family, (B) LEBRE ÷ core over all series with the upper limit of the 95% CI below 1 and (C) estimated memory ≤ 128 KB.

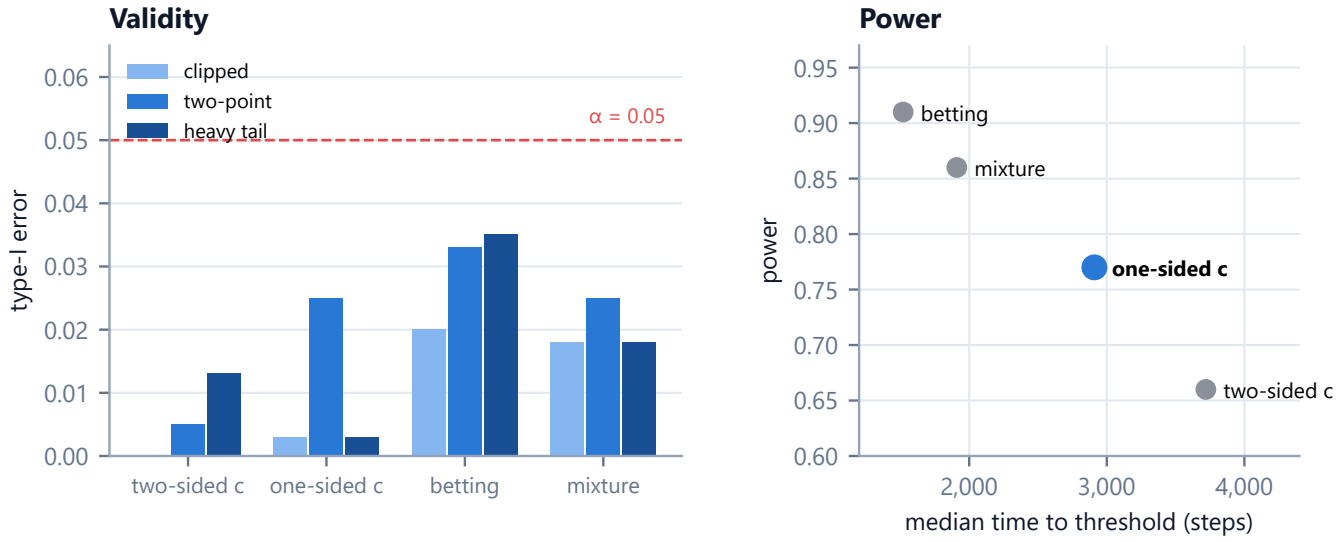
(c) v0.54 against the previous version

The data of the v0.54 final evaluation were reserved on 2026-10-10, before any code of this version: 93 series (30 basins, 30 buildings, 8 wind plants, 8 solar plants in Jan–Sep 2026, 17 levels of EIA prices and US Treasury rates). The ruler is the same as in (b), with the previous version (core + M1 + M2) in the role of the core: (A1) on "reference better" series, v0.54 ÷ R with upper limit ≤ 1.01; (A2) on "previous version better", v0.54 ÷ previous version with upper limit ≤ 1.02; (A3) on levels, no additional series with an accepted external input; (A4) short series ≤ 1.05; (A5) no non-finite forecast. **Promotion:** A, plus (B) median per-series cost ratio with the upper limit of the 95% CI below 1 and (C) estimated memory ≤ 128 KB. Each change of v0.54 was first measured on the development bench and, after revision, validated on new data, used once (§17, §18).

15. Evidence for the structural core

The core was developed and evaluated before M1 and M2 existed, with protocol (a) of §14. Since M1 and M2 do not alter any decision of the core (identical structure verified on every series of development, validations and the previous final evaluation), this evidence still holds for the core within LEBRE. Two changes of v0.54 lie inside the core: the cheap equivalence report changes no forecast nor decision; the triggered removal changes the trajectories only when it opens a test. While no triggered test opens, the core is identical to the one evaluated in this section (verified in the identity tests of development and of the audit).

15.1 Development: validity and discovery



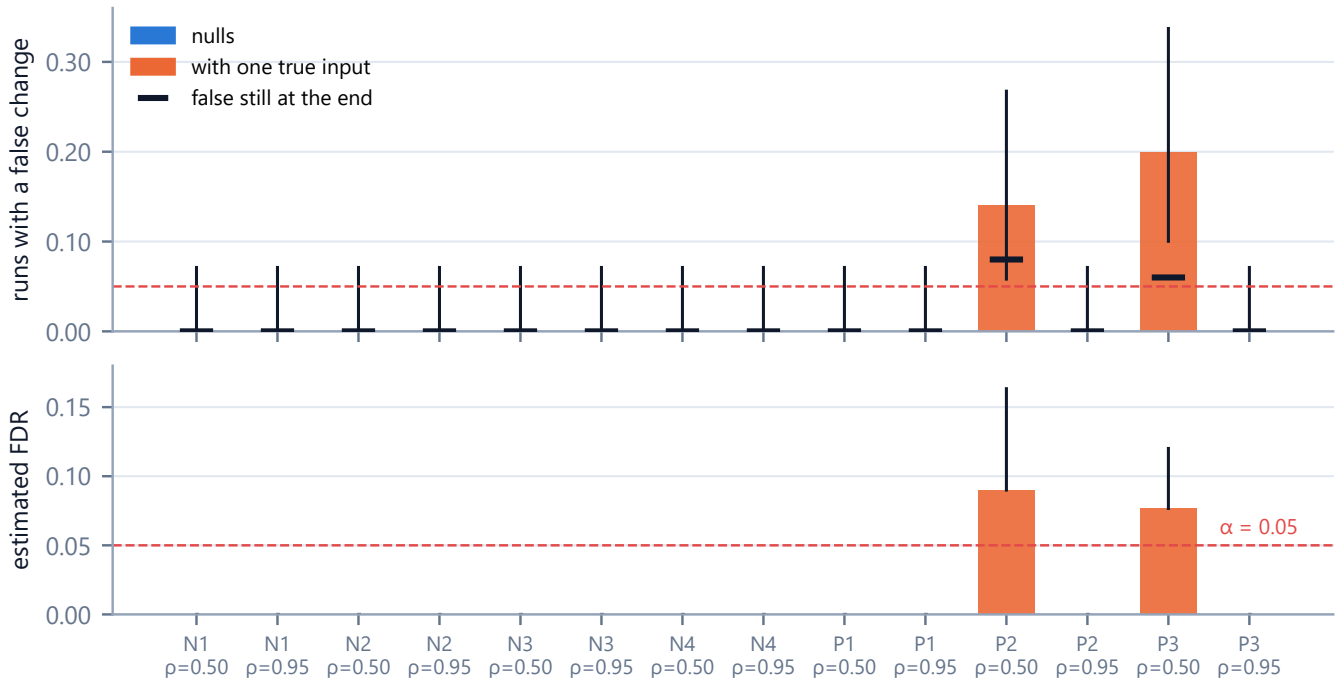
E-process simulation (400 trajectories × 5,000 steps). Left: type-I error under three increment shapes (clipped, two-point, heavy tail) at the null boundary, against α = 0.05. Right: power and median time to the threshold for an improvement of +0.0072 per sample. The adopted variant is the one-sided c.

Situation	Result (development)
Synthetic with white inputs (delays up to 12)	structure recovered to the exact delay in 500–2,200 steps (e.g. delay 12, weight 0.84 vs 0.80)
Semi-synthetic (real inputs, known effect)	true input (or its group) in 2 of 3 structured tasks, 3/3 seeds; latency 4–17 thousand steps
Very smooth inputs (autocorrelation ~0.99), medium/long delay	**not discovered**; an online oracle (knows the delay, tests from the start) needs 9–27 thousand steps or never gets there in 32 thousand
Structureless targets (null batches with real inputs)	0/40 and 0/30 acceptances; 0 false exogenous inputs in all declared sets
Near-identical inputs (correlation 0.98)	indistinguishable by predictive improvement; the model reports the group

The limit on smooth inputs is known in principle: with very persistent regressors, the new information of a delay beyond the current value is small (Stambaugh 1999; Campbell & Yogo 2006). In those cases the forecast remains good because the base already uses the current input values; what is lost is the *explanation* of the delay.

15.2 False-change control in simulation with dependence

The frozen core was run in 700 synthetic runs (plan and criterion declared beforehand; 4 mutually correlated inputs, persistence 0.5 or 0.95, 30 thousand steps): four nulls (white noise; autoregressive target; noise whose variance depends on an input; variance break) and three scenarios with one true input at delay 6 (plain; with a decoy of correlation 0.7 with the true one; with heavy-tailed autoregressive noise).



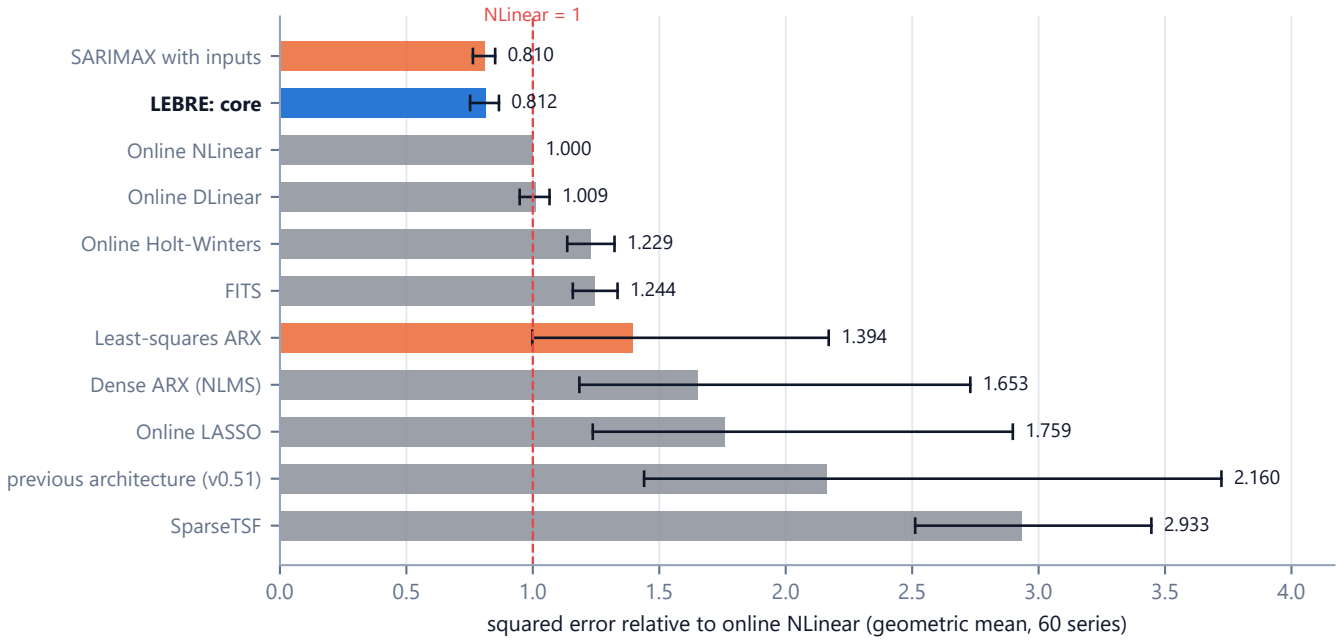
Top: fraction of runs with at least one false change (Clopper-Pearson 95% CI) and fraction where the false change was still in the model at the end (dash). Bottom: estimated false change rate (FDR) with bootstrap upper bound. Blue: nulls; orange: scenarios with one true input.

- **Nulls:** 0 of 400 runs with any accepted change.
- **With one true input:** by the declared criterion (per scenario, 100 runs) the three scenarios stayed within α = 0.05 (FDR 0.045 and 0.038 in the worst). But, with persistence 0.5, two cells exceeded 0.05: FDR 0.090 (decoy) and 0.077 (dependent noise), with 7/50 and 10/50 runs affected.
- **Mechanism:** in all 17 affected runs, the false change was an input *correlated* with the true one, accepted before it. While the true input is not in the model, this substitute does improve the reference forecast — the test answers its own question correctly, but the answer is not structural. The periodic removal test took the substitute out in 10 of the 17 cases; in 7 of 700 runs it stayed in the final model.

What this allows us to claim: in the simulated scenarios, the procedure empirically controlled false *predictive improvements* and accepted nothing on structureless targets. It does **not** guarantee true structure: with correlated inputs, a substitute may be accepted and sometimes remain. An accepted input means "it improved the forecast of the current reference, with evidence", not "it is the cause".

15.3 Evaluations on never-seen series

Reserve 1 (125 series): an engineering failure. The first pre-registered evaluation, with the core still without the output safeguards, revealed that **6 of 125 series exploded** (relative error from 13 up to 10^{79}): when the target was missing for many steps, the memory filled its own history with its own forecasts. The pre-registered result was dominated by these series (geometric mean 5.20); in an exploratory view, not pre-registered, without the 6 series, the relative error was 0.706, equal to development. The failure motivated the safeguards of §11, checked in development and then on never-seen series.



Reserve 3 (60 never-seen series). Geometric mean of the squared error relative to online NLinear, with 95% CI. Blue: LEBRE core; grey: other budget-class models; orange: classical references.

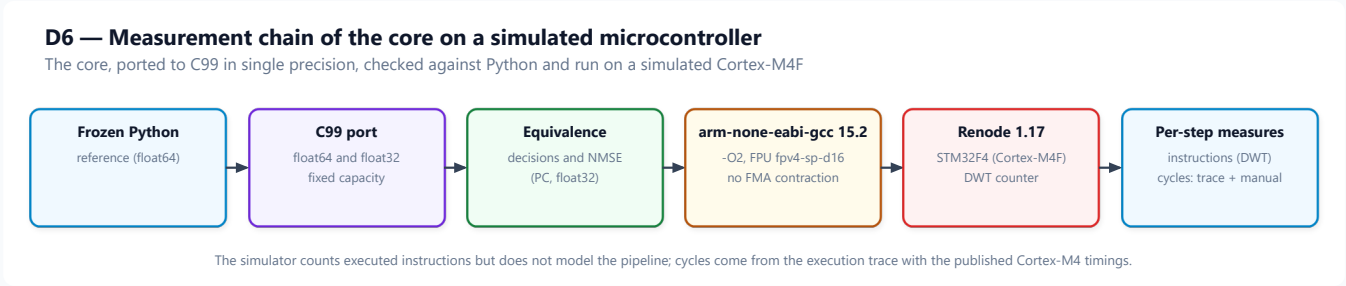
Model	CAMELS	BDG2	All [95% CI]	Series > 1.5	Catastrophic	Operations/forecast
LEBRE: core	**0.858**	**0.768**	**0.812 [0.752; 0.866]**	**0%**	**0**	**415**
Online NLinear	1.000	1.000	1.000 [1.000; 1.000]	0%	0	674
Online DLinear	1.076	0.947	1.009 [0.948; 1.066]	3%	0	1,533
Online Holt-Winters	1.213	1.244	1.229 [1.136; 1.323]	25%	0	20
FITS	1.362	1.137	1.244 [1.158; 1.335]	18%	0	9,757
Dense ARX (NLMS)	1.008	2.711	1.653 [1.184; 2.730]	38%	2	474
Online LASSO	1.015	3.048	1.759 [1.237; 2.898]	38%	4	705
previous architecture (v0.51)	2.180	2.140	2.160 [1.440; 3.723]	48%	4	108
SparseTSF	2.980	2.886	2.933 [2.512; 3.446]	87%	3	4,016
SARIMAX with inputs	0.817	0.803	0.810 [0.763; 0.851]	0%	0	ML fit
Least-squares ARX	0.677	2.867	1.394 [0.998; 2.170]	20%	4	48,944

Ceiling (1,000 points)	CAMELS	BDG2	All	Paired ratio core/model [95% CI]	Operations/forecast
SARIMAX with inputs	0.798	0.818	0.808	1.004 [0.956; 1.056]	ML fit
TTM zero-shot	1.204	0.978	1.085	0.747 [0.704; 0.795]	$15.6 \cdot 10^6$
TTM fine-tuned w/ inputs	1.049	1.054	1.052	0.771 [0.699; 0.844]	$55.6 \cdot 10^6$
Chronos-2 w/ covariates	0.782	0.539	0.649	1.250 [1.178; 1.337]	$\sim 10^9$–10^{10}

Summary of the core evaluations with the safeguards: error relative to NLinear of 0.767 (reserve 2, 40 series) and 0.812 (reserve 3, 60 series); no catastrophic series and none above 1.5 among the 100 series; a tie with SARIMAX with inputs in both (paired ratios 0.999 and 1.002); lower error than TTM zero-shot and fine-tuned; error 16% and 25% above Chronos-2 with covariates. The ultralight FITS and SparseTSF, batch-fitted on the first 15%, were behind online NLinear in this one-step regime, for which they were not designed. The 90% interval had empirical coverage of 0.905 and 0.904.

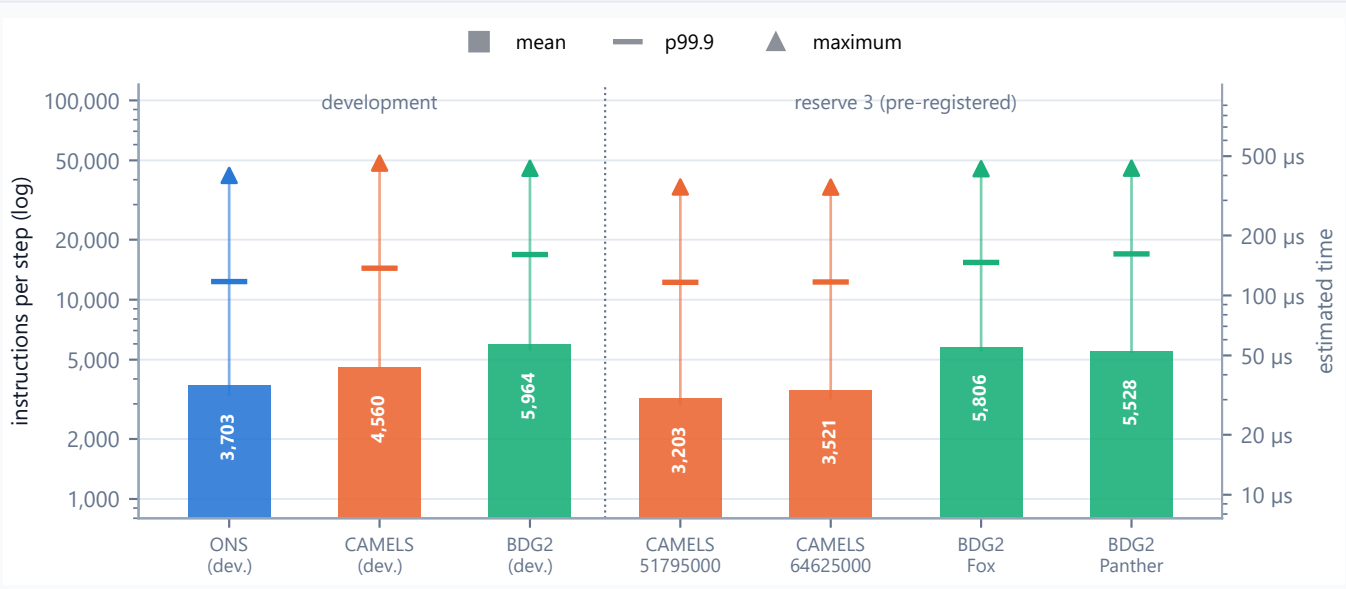
16. The core on a simulated microcontroller

To check whether the analytic cost corresponds to something executable on a microcontroller, the frozen core was ported to C99 and run on a **simulator** of a common microcontroller (STM32F4, Cortex-M4F core at 168 MHz). There was no measurement on a physical board. **M1 and M2 were not ported**: for them there is only the analytic count (§12).



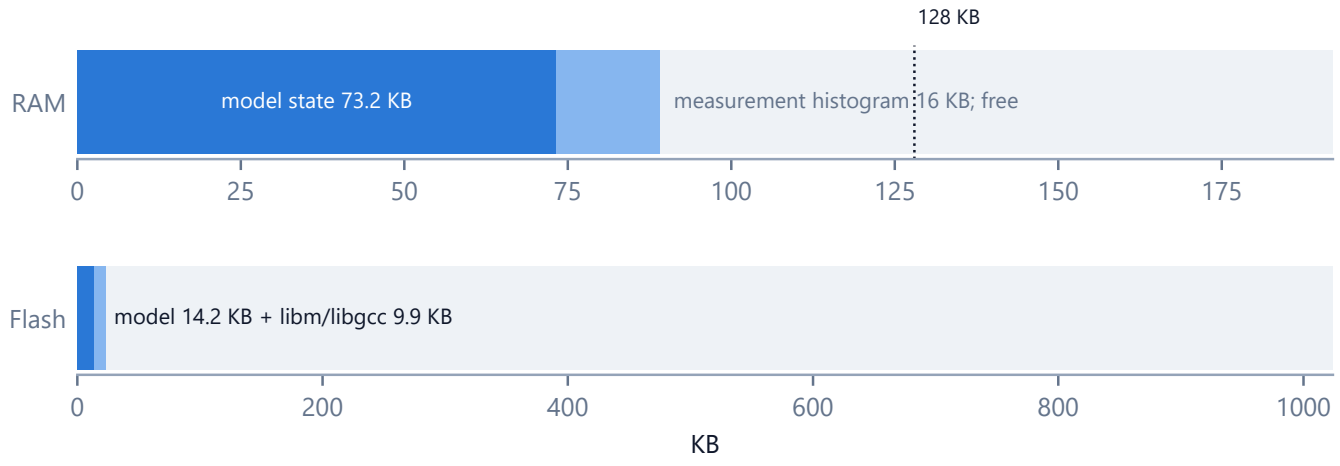
Measurement chain. Equivalence was checked first on the PC (C in double and single precision against Python), then on the simulated device.

Equivalence. In double precision, the C code exactly reproduces the Python decisions on 29/29 development series. In **single precision** (the microcontroller build), the sequences of accepted changes are identical on 28/29 development series and on 54/60 never-seen series; on the others, a decision near the threshold happens a few steps earlier or later (rounding), with a median error difference of $\sim 10^{-6}$. **Cost.** The simulator (Renode 1.17) counts executed instructions via the DWT counter, but does not model the pipeline; cycles were estimated from execution traces with the published Cortex-M4 and FPU timings: CPI 1.62 and 1.57. Times below use CPI 1.6 and are estimates.



Core instructions per step on 7 series (3 development, 4 from the pre-registered reserve 3): mean (bar), p99.9 (dash) and maximum (triangle). Right axis: estimated time at 168 MHz.

Series	Source	Inputs	Mean	Median	p99.9	Peak	Time	NMSE	Changes
ONS (dev.)	dev.	1	3,703	3,363	12,327	42,021	~35 µs	0.2826	4
CAMELS (dev.)	dev.	3	4,560	4,599	14,387	48,447	~43 µs	0.0229	3
BDG2 (dev.)	dev.	2	5,964	5,607	16,843	45,654	~57 µs	0.0305	2
CAMELS 51795000	reserve 3	3	3,203	3,003	12,223	36,771	~31 µs	0.1465	0
CAMELS 64625000	reserve 3	3	3,521	3,215	12,267	36,708	~34 µs	0.3249	0
BDG2 Fox education Ollie	reserve 3	2	5,806	5,587	15,375	45,402	~55 µs	0.0162	3
BDG2 Panther education Vincent	reserve 3	2	5,528	5,463	16,971	45,696	~53 µs	0.0584	3



Memory on the STM32F407 (192 KB RAM, 1 MB flash). RAM: core state and other variables of the measurement firmware. Flash: core and math-library code.

Reading: ~3.2–6.0 thousand instructions per step correspond to ~9–15 instructions per analytic operation (memory accesses, loops, indices, divisions). At ~31–57 µs per step, a 1-second sampling would take < 0.01% of the processor core time; energy was not measured. With M1 and M2, which roughly double the analytic count (§12), the expected order of magnitude stays the same, but it **was not measured**. The state fits the STM32F407 RAM, but not smaller microcontrollers without reducing capacities.

17. Development of the layers and of v0.54

A public test bench (the LEBRE Lab) ran the frozen core on new domains and surfaced failure modes. M1 and M2 address two of them, without widening the core's hypothesis space; the others were left outside this version (§22).

Failure mode 1: losing to the trivial reference (motivates M2)

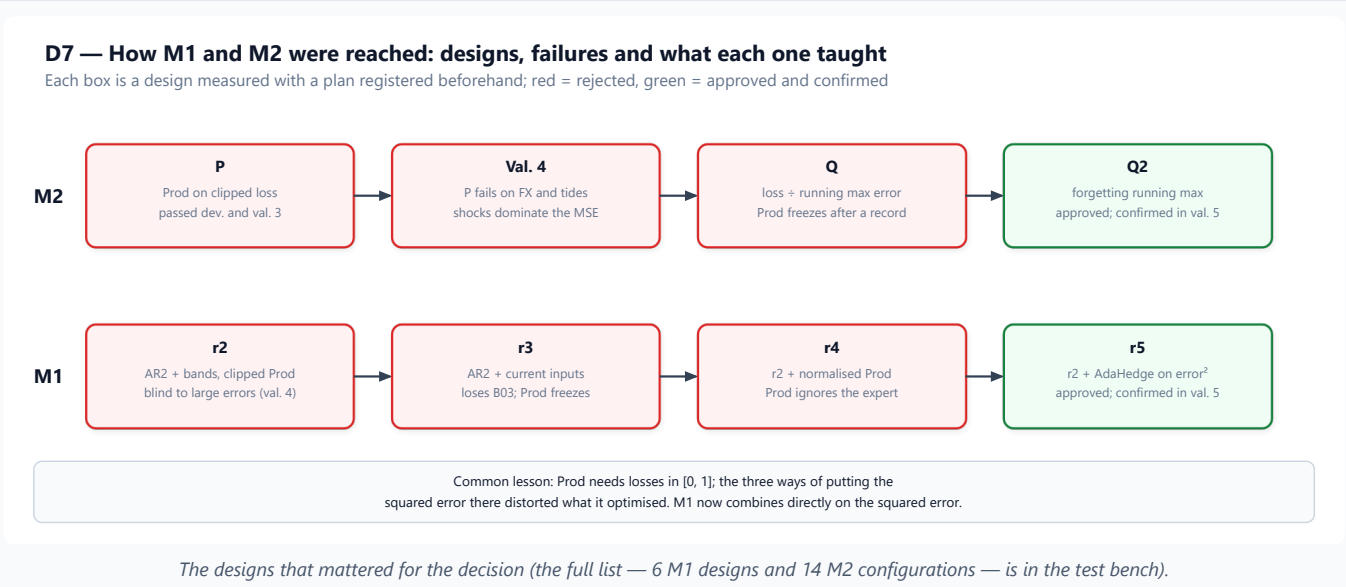
In nearly unpredictable series, the core's adaptive part errs more than the trivial reference. In the final reserve itself, the reference beat the core, with the whole confidence interval above 1, in 5 of 8 FX return series and 6 of 8 level series.

Failure mode 2: below the accuracy × cost frontier (motivates M1)

In 8 development families a full linear comparator (all inputs and lags estimated jointly, from thousands to tens of thousands of operations per step) beats the core by more than 5%. What matters under a budget is whether the core loses to something *of the same cost*. The diagnostic measured 8 regressions of varied cost in each family: the core is below the frontier (a regression of equal or lower cost errs 5% less) in 4 of them, all with the same cause, the lack of joint estimation of the target dynamics and of the input effect.

Family	core ÷ full	best with cost ≤ core	core cost (FP)	status
B03	2.076	1.574	731	**below**
VB07	1.170	1.005	435	**below**
XB07	1.141	1.010	417	**below**
B07	1.067	1.004	431	**below**
YB06	1.095	1.048	640	on the frontier
YB03	1.225	1.397	830	on the frontier
VC05	1.138	1.381	262	on the frontier
A05	1.077	1.777	385	on the frontier

Error = MSE ÷ MSE of the full online linear comparator (family geometric mean). B03: air quality; B07, VB07, XB07: storm surge (three independent sets); VC05: inflation; YB03: Beijing air quality; YB06: wind; A05: synthetic.



The designs that mattered for the decision (the full list — 6 M1 designs and 14 M2 configurations — is in the test bench).

The development bench grew to 64 families and 728 series (the original development, validations 1 to 4, already spent, and the short families). On it, the final combination (M1 draft 5 + M2 Q2) met all M2 criteria and F6 on the four families below the frontier:

Family	core ÷ full	LEBRE ÷ full	target	cost ÷ core
B03	2.076	1.365	1.652	2.01
B07	1.067	1.020	1.054	2.17
VB07	1.170	1.053	1.056	2.13
XB07	1.141	1.044	1.061	2.21

Process findings, all recorded: a numerical defect in path W (0/0 by *underflow*) was found in a test, fixed with the log-scale fallback and audited (it never fired in a recorded measurement; since then a non-finite forecast counts as a failure); a construction error in validations 4 and 5 (30 h cycle instead of 24 h in solar, wind and load) restricted the scope of validation 5 (§18).

Development of v0.54

The baseline measured before this version showed three problems beyond those addressed by M1 and M2: (1) M1 worked with zero weight on many series and cost on all of them; (2) the core cost grew with never-accepted inputs; (3) removals never happened — after a regime change, the old input stayed in the structure in 20 of 20 series. Also examined were (4) spurious inputs in levels and (5) the cycle that must be declared. Each change was measured on the 64-family bench against the previous configuration, with the plan committed beforehand; a design revised after a measurement was only accepted after passing on new data.

Change	Measured designs	What happened	Status
Dormant M1	3	M1 cost in many-input scenarios: 0.570, 0.264 and 0.373 of the previous one; the 2nd design hurt air quality (+5% on one series), the 3rd met everything but the 75% cut target, revised to 60% by the project owner and declared	accepted (validation 6)
Cheap core with many inputs	3	dormancy of inputs in the screen discarded before measuring (it delayed or prevented the discovery of weak inputs); the 1st design of the cheap report had *aliasing* with the 24 h cycle; the 2nd passed; the audit found the cost peak (errata 01)	accepted (validation 7 and errata 01)
Removals that happen	2	the 1st design removed intermittent relations in the off blocks (+14% error, upper limit 1.145) and changed the test schedule; the 2nd, added to the schedule and with a minimum duration, had a worst upper limit of 1.006	accepted (validation 7)
Spurious inputs in levels	0	diagnosis: the tests compare against the structural forecast, which does not start from the last value; in a random walk, any input that also wanders seems to help; removal does not solve it, because its contribution does not drop	outside this version
Cycle as a hypothesis	2	the evidence had to be the per-phase profile (lag features give false cycles in AR series); without declaring the cycle, load reached 1.04 times the version with the declared cycle, but solar stayed at 1.20 (late acceptance); no false cycle in series without a cycle	outside this version

The numbers of each measurement, with plans and results, are in the test bench.

18. Validations on new data

Validations 1 and 2 served earlier M2 designs; validation 3 confirmed candidate P. **Validation 4** rejected M1 draft 2 with P: criterion 1 on YB07, YC01 and criterion 2 on YB01. The diagnostic separated the causes (the solar failure came from M1; the FX and tide ones from M2) and led to the designs of §17.

Validation 5 (43 new series, single measurement) confirmed M1 draft 5 with M2 Q2: F6 met on Beijing air quality (Chen, 2017) and on six new tide stations, and all M2 criteria on every family.

Family	core ÷ full	LEBRE ÷ full	target	cost ÷ core
Z-B03	1.169	1.118	1.410	1.97
Z-B07	1.017	0.925	1.068	2.20

> **Errata:** in validations 4 and 5, the solar, wind and load families were built with a 30 h cycle instead of 24 h. The core vs LEBRE comparisons in those families hold for the task as built, not for the declared analogue; the fresh evidence for LEBRE with the right cycle in those domains is the previous final evaluation (§19.4).

Validations of v0.54

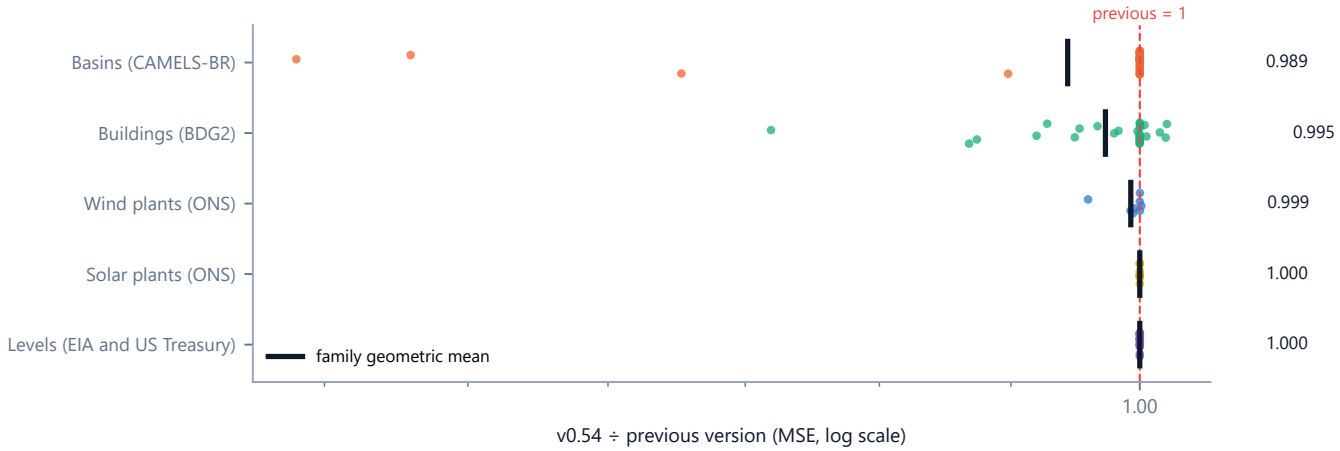
- **Validation 6** (dormant M1; 70 never-seen series: 20 real ones of load, air quality, wind and tide and 50 synthetic): M1 cost ratio 0.377 and 0.229 on the cost scenarios; no family with an upper limit above 1.01 (worst 1.006). Accepted.

- **Validation 7** (cheap report and triggered removal; 100 never-seen series: 90 synthetic, 4 air-quality ones from a new station and 6 never-used wind plants): identical report and identical forecasts; core cost with 50 inputs 0.719 of the previous one; the old input removed after the regime change in 20 of 20 series; no removal of a true input; worst upper limit 1.005. Both accepted; on 2 series with an intermittent relation, the removal happened when the relation returned (declared).
- **Errata 01** (report spread over rows): on the bench, identity with the promoted version on 728 of 728 series and peak fixed, but not accepted by the wording of the cost criterion (on 70 series, the difference was exactly the cost of the rows still pending at the end of the series); with the criterion rewritten and declared, accepted on 60 new series, with a peak of at most 1.054 times that of the previous version.

19. Pre-registered final evaluation, audit and errata

19.1 The final evaluation of v0.54

Single run, with ruler (c) of §14, on 93 series reserved before any code of this version: 30 CAMELS-BR basins, 30 BDG2 meters, 8 wind and 8 solar ONS plants (solar in Jan–Sep 2026, a time split) and 17 daily levels (10 EIA spot prices and 7 US Treasury rates, with WTI, Brent and the 2- and 10-year nominal rates as inputs). The pre-registration was written before the access and went through a rehearsal on already spent series. Result: **promoted** by the rule fixed beforehand.



v0.54 ÷ previous version per series (dots) and family geometric mean (bar).

Family	series	previous better / ref. / undec. / shock	v0.54 ÷ previous	A1	A2	cost (median ratio)	removals
Basins (CAMELS-BR)	30	16/0/3/11	0.989	-	0.992 [0.989; 0.994]	1.006	0
Buildings (BDG2)	30	30/0/0/0	0.995	-	0.995 [0.993; 0.997]	0.847	1
Wind plants (ONS)	8	8/0/0/0	0.999	-	0.999 [0.997; 1.000]	0.753	0
Solar plants (ONS)	8	8/0/0/0	1.000	-	1.000 [1.000; 1.000]	0.917	0
Levels (EIA and US Treasury)	17	0/2/15/0	1.000	1.000 [1.000; 1.000]	-	0.879	0

- Cost (B): **0.905** [0.857; 0.932]; short series against the best of the previous version and R: 1.007 [0.998; 1.008]; estimated memory 85,268 B.
- No non-finite forecast; on levels, external inputs accepted on 3 series, as in the previous version (3) — M4 does not solve the interpretation in levels (§22); M1 slept on 64 of the 93 series; over the whole reserve there was 1 accepted removal, from a periodic test (which already existed), and no triggered removal; the removed unit came back later, and the final structure is the previous version's.
- An infrastructure failure in the first pass (the parallel processes imported the previous version's data reader, which has the same name) was fixed with no model or criterion change; 4 results computed with wrong data (2024–2025 instead of 2026, already spent) were removed before any analysis. Chronos-2 was completed after an environment failure, and SARIMAX with inputs did not run on levels (inputs with gaps), as with FX levels before.

19.2 Reference ceilings (reported only)

Models outside the budget class, as reference only: Chronos-2 with covariates (Ansari et al., 2025), on the same 1,000 points per series as protocol (a); and SARIMAX with inputs, fitted on the first 15%.

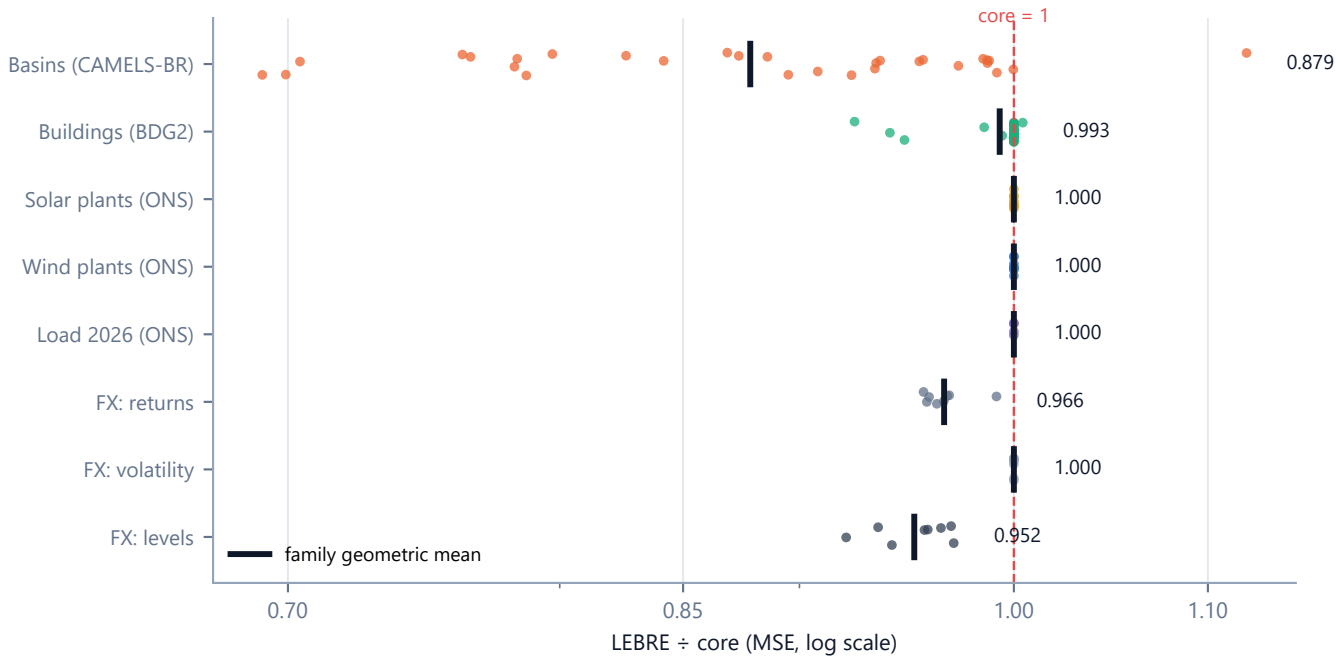
Family	v0.54 ÷ Chronos-2	previous ÷ Chronos-2	SARIMAX-X ÷ v0.54
Basins (CAMELS-BR)	0.943	0.962	1.089
Buildings (BDG2)	1.338	1.342	1.042
Wind plants (ONS)	1.060	1.061	1.057
Solar plants (ONS)	1.319	1.319	1.106
Levels (EIA and US Treasury)	0.835	0.835	–

19.3 What the evaluation does not cover

There is no known truth about regime changes in the real series: the evaluation checks that removing does no harm, and the evidence that removal happens when it should comes from development and validations. The reserve had no electric load nor FX (exhausted) nor long series with relations that disappear and return.

19.4 The previous final evaluation: M1 and M2 on top of the core

The previous version (core + M1 + M2, without the v0.54 mechanisms) had its own pre-registered final evaluation, which is the evidence that M1 and M2 improve the core. Single run on 104 reserved series (30 CAMELS-BR basins; 30 BDG2 meters; 8 solar plants, 8 wind plants, the 2026 load of the 4 ONS subsystems and 8 FX pairs in 3 tasks: returns, volatility and levels). The pre-registration was approved after a rehearsal on 55 already spent basin and building series (0 errors). Result: **promoted** by the rule fixed beforehand.



LEBRE ÷ core per series (dots) and family geometric mean (bar).

Family	series	core better / ref. / undec. / shock	LEBRE ÷ core	crit. 1	crit. 2	max +cost (FP)
Basins (CAMELS-BR)	30	15/2/7/6	0.879	0.845 [0.814; 0.878]	0.911 [0.899; 0.924]	495
Buildings (BDG2)	30	30/0/0/0	0.993	-	0.993 [0.988; 1.000]	385
Solar plants (ONS)	8	8/0/0/0	1.000	-	1.000 [1.000; 1.000]	696
Wind plants (ONS)	8	8/0/0/0	1.000	-	1.000 [1.000; 1.000]	918
Load 2026 (ONS)	4	4/0/0/0	1.000	-	1.000 [1.000; 1.000]	831
FX: returns	8	0/5/1/2	0.966	1.000 [1.000; 1.000]	-	894
FX: volatility	8	6/0/0/2	1.000	-	1.000 [1.000; 1.000]	894
FX: levels	8	0/6/0/2	0.952	1.001 [1.000; 1.002]	-	894

- Overall improvement: **0.955 [0.951; 0.960]**; short series against the best of core and R: 1.007 [0.995; 1.017].
- No non-finite forecast; false inputs on returns equal to the core's; structure preserved on every series; estimated memory 85,092 B.
- In solar, wind and load, M1 ended with zero weight on all 20 series and LEBRE was identical to the core.

- Two fixes during the run, with no effect on model or criterion: reading the 2026 load by subsystem code (ONS renamed the Southeast) and handling a degenerate short series (exact tie with zero error).

Reference ceilings in that evaluation (reported only):

Family	LEBRE ÷ Chronos-2	core ÷ Chronos-2	SARIMAX-X ÷ LEBRE
Basins (CAMELS-BR)	0.885	1.008	1.134
Buildings (BDG2)	1.248	1.241	1.034
Solar plants (ONS)	1.237	1.237	1.096
Wind plants (ONS)	1.051	1.051	1.049
Load 2026 (ONS)	1.433	1.433	0.968
FX: returns	0.954	0.979	1.123*
FX: volatility	0.983	0.983	1.058*
FX: levels	0.881	0.921	1.065*

* FX: SARIMAX with inputs did not run in the execution (inputs with missing holidays); values from an addendum made after the result, with inputs filled by the last value.

19.5 Stress audit after the promotion and errata 01

At the project owner's request, the promoted v0.54 went through an audit with a plan and alert signals fixed beforehand, which does not redo the promotion: it looks for defects the evaluation did not cover.

Test	Result
Identity and determinism (6 cases)	with the mechanisms off, identical to the previous version; deterministic; single step equal to predict + observe
Robustness (12 stress types)	no non-finite forecast; error ratios from 0.974 to 1.005 (the constant target gave MSE ~10^{-32}, rounding)
Nulls (100 series)	external inputs in the structure: 0, as in the previous version (0)
Weak true inputs (80 series)	no removal of a true input; ratios from 0.982 to 1.001
M1 wake-up (40 series)	ratios after the change from 0.997 to 1.003
State (up to 200 thousand steps)	no new growing container; the event list grows without bound, as already in the previous version
Whole combination on the bench (64 families)	no upper limit above 1.02 (worst: YB03, 1.007)
Intermittency	**alert:** with on/off blocks of 7,500, 10,000 and 15,000 steps, error 1.100, 1.055 and 1.046 times that of the previous version; up to 5,000 steps, no effect
Peak cost	**alert:** per-step peak up to 1.522 times that of the previous version with 50 inputs; mean cost lower in almost all groups and wall-clock time 0.845

Errata 01. (1) The peak came from the whole update of the equivalence report coinciding with M1's regression; the fix applies the same sample row by row, with identical forecasts, decisions, report and total cost, and the peak came to at most 1.054 times (validated on new data, §18). The forecasts evaluated in §19.1 are therefore those of the corrected version. (2) The removal of relations that disappear for more than ~5,200 steps and return is declared as a limit (§22): the removal is valid for the episode, but LEBRE keeps no memory that the relation existed and must re-detect it (median of ~700 to ~900 steps).

20. Evolution of the architecture and ablations

Evolution

LEBRE evolved through frozen versions, each with its hashes. v0.1 (the project's canonical reference) defined the idea: structure that enters and leaves the model by evidence, within a budget. v0.51 used two experts with per-lag discovery and martingale/CUSUM tests; later analyses showed that, on real series, almost all of the result came from the memory. v0.52 kept the memory and the combination and replaced discovery by a generic engine of tested changes — it is this document's **structural core**, frozen on 2026-09-27 without promotion. v0.53 added M1 and M2 without changing a byte of the core and was promoted on 2026-10-09. v0.54 (this document) added M1 dormancy, the cheap equivalence report and the triggered removal, and was promoted on 2026-10-13 (with errata 01 of 2026-10-14). The internal steps of the core were:

1. per-lag hypotheses (~400) → **per whole input** (fewer hypotheses; better on real data);
2. bands diluted isolated delays → **peak lag** and hierarchical splits;
3. slow evidence → **one-sided c** and persistent peak statistic;
4. acceptances on flat stretches → loss **scale floor**;
5. acceptances on structureless targets → **step 0.05** in the reference (§8);
6. cost peaks → one warm-up at a time, spread screen, half-window peak statistic;
7. explosions in gaps → **hold and output contract** (§11).

Ablations

Variant	What changes	Result
Core alone (no M1 and M2)	the output is L	previous final evaluation: LEBRE ÷ core = 0.955 [0.951; 0.960]; M1 has zero weight in solar, wind and load
Core: all on, no tests	all inputs and the latent state active from the start	reserves 2 and 3 (100 series): ratio 0.988 [0.970; 1.006], ~5% lower cost — exploratory; **the testing mechanism does not improve accuracy** distinguishably
Core without hierarchical splits	the response stays at octave bands + peak	ratio 0.996 [0.988; 1.001] — no difference on real data
Per-lag hypothesis ("atomic")	one hypothesis per lag of each input (~400 hypotheses)	development: 0.757 vs 0.645 (1,000 points); slower discovery
No output safeguards	open-loop memory during gaps	reserve 2: 1.107 vs 0.767; 2 catastrophic; paired ratio 0.693
Base with step 0.1	larger reference misadjustment	10/40 acceptances on structureless targets (§8)
Previous architecture (v0.51)	two experts with per-lag discovery, no stream control	reserve 3: 2.160, 4 catastrophic
M1 and M2 with other combiners	clipped Prod, normalised Prod, other designs (§17)	rejected in development or in validation 4
No M1 dormancy	M1 runs fully at every step	validation 6: same accuracy (worst upper limit 1.006), M1 cost 2.7 times higher in the many-input scenario
No triggered removal	only the periodic schedule with a persistent hypothesis	validation 7: after the regime change, the old input is not removed (0 of 20 at the baseline; 20 of 20 with it)
Equivalence report every 4 steps	whole covariance matrix every 4 steps	identical forecasts; core cost with 50 inputs 1.39 times higher

Consequence for what can be claimed: the core's accuracy comes mainly from a cheap online linear regression on the current inputs and a seasonal memory, protected by output safeguards; LEBRE's gain over the core comes from M1 and M2 (§19). The mechanism of tested changes **is not justified by an accuracy gain**; it is justified, if at all, by the structure *logged with evidence* (which inputs, with which delay, since when, with which evidence). **We did not demonstrate** that this structure improves any practical decision (diagnosis, control, maintenance).

21. Position in the literature

Aspect	Related work	Classification
Structural change accepted by an e-process of predictive improvement, with stream error control	Amoukou, Mishra & Veloso (2026), in online decision trees (FWER control); Choi (2026), prespecified predictive corrections; Jha (2026), Granger causality with sequential testing; Choe & Ramdas (2024); Xu & Ramdas (2024)	**not original**
Per-input units, hierarchical refinement, tested removal and swap, FDR control, fixed budget	known parts (hierarchical testing; e-LOND; persistent hypotheses with FDR: de Heide 2026; doubly sequential testing: Yao, Gang & Sun 2025)	integration/extension
Aggregation of forecasters (M1, M2)	AdaHedge (de Rooij et al. 2014); (A,B)-Prod and constant regret against a trusted expert (Sani et al. 2014; Even-Dar et al. 2008; Koolen 2013); switch distribution (van Erven et al. 2012) and Fixed Share (Adamskiy et al. 2016; Cesa-Bianchi et al. 2012); recursive regression with lags (Ljung 1999)	**not original**; choice of losses and chaining
Unbounded and heavy-tailed losses	Cesa-Bianchi et al. (2007); Hazan & Kale (2010); Orabona & Pál (2018); Mhammedi et al. (2019); V'yugin & Trunov (2019); Alquier (2021); Moulin et al. (2025); robust confidence sequences (Wang & Ramdas 2023), future direction	known limit; handled in the ruler
Condition of use (reference misadjustment)	Giacomini & White (2006); Clark & West (2007); LMS misadjustment (Widrow & Stearns 1985); Han & Qu (2026); incumbent bias in challenger promotion (Fernández-Barrios et al. 2026)	modest methodological observation
Forecasting with a budget of ~10^2–10^3 operations per step	SparseTSF, FITS, TTM (generally univariate and batch-trained); embedded forecasting with incremental learning and exogenous sensors (Lara Pinal, Das & Schlüter 2026)	empirical contribution, within the tested scope
Hold, output contract, cost limits	engineering practice	standard

The literature audit of the core followed a protocol with claims and queries fixed beforehand (47 queries, in two rounds) and classified each work as anticipation, partial, basis or distant; no work found anticipates the whole combination, and each part has a precedent. **Limitations:** one general search engine, English queries, reading mostly by abstract, no double screening; it is not a systematic review in the formal sense. LEBRE's contribution is the **integration** of these components under a declared budget and the pre-registered evaluation.

22. Limits and claims we do not make

- **Horizon:** one step ahead only, with the current inputs available.
- **Domains:** the core was evaluated against the budget class on hydrology and buildings; M1 and M2, on basins, buildings, solar, wind, load and FX, plus air quality and tides in the validations. Other domains were not tested.
- **Cost without gain in part of the domains:** where M1 does not help, it now sleeps and costs little, but M2 and the dormant M1 still cost on every series; the many-input cost target (≤ 2 times the cost with 5) was not met: the cheap report only cuts the quadratic part.
- **Relations that disappear and return:** a relation absent for more than ~5,200 steps is removed and must be re-detected when it returns; with blocks of 7,500 to 15,000 steps, the error is 1.046 to 1.100 times that of the previous version (§19.5). Seasonal relations in hourly data may have that shape.
- **Interpretation in levels:** spurious external inputs are still accepted in level series (accuracy is protected by M2, but the reported structure is not). The mechanism was diagnosed: the tests compare against the structural forecast, which does not start from the last value.
- **Declared cycle:** the cycle still needs to be declared; treating it as a hypothesis did not meet the target on solar.
- **Event list:** grows without bound (already in the previous version); on an embedded device, it must be bounded.
- **Core cost contract** not met (3.7% in the mean and 2.0% in the peak), without later revision; the M1 + M2 ceiling was redefined before the previous final evaluation (§12).
- **Core testing mechanism:** does not improve accuracy; false-change control is in the practical regime, not in the exact-guarantee one (§8); inputs correlated with the true one may be accepted as substitutes and sometimes remain (§15).
- **Discovery:** fails on very smooth inputs with medium/long delays; near-identical inputs are indistinguishable.
- **No formal guarantee in the original scale for M1 and M2:** Prod guarantees on normalised losses; M1's AdaHedge gives no constant regret against the core; with shocks, no combiner has a useful guarantee (§6.4).
- **Ruler decided during development:** the shock addenda were decided after seeing development data, before validation 5 and the previous final evaluation.
- **F6 without a family in the final reserve:** the evidence for F6 on new data is validation 5 only; one development family passed by a small margin (VB07).
- **Microcontroller:** core only, simulator only, cycles estimated, energy not measured; M1 and M2 memory estimated.
- **Uncertainty:** the overall CIs treat series as independent; load has 4 series; 4 BDG2 sites share weather with the core's development.
- **Missing comparators:** an online ARMAX with a Kalman filter and a SARIMAX with online parameter updates.
- **Interval:** empirical coverage near 90% in the core, without a formal guarantee; not reported for the complete output.
- **Post-failure fixes:** the safeguards and the final M1 and M2 designs came after seeing failures; they were confirmed on new data, but the design was not blind. Errata and run fixes are declared.
- **Revised targets and rewritten criteria:** M1's cost-cut target was revised (from 75% to 60%) after development measurements and the cost criterion of errata 01 was rewritten after the first measurement; in both cases acceptance required a new validation on never-seen data, and the revisions are declared.
- **Outside this version:** interpretation in levels, the cycle as a hypothesis and multiple outputs or multiple steps.

> **We do not claim** that LEBRE is the best forecaster, nor the most economical, nor that any component is new. We only claim what was measured: under the protocols and data described, at the cost described, the numbers reported in §15–§19.

23. Parameters

All constants are the same for any series; nothing is tuned per series.

Core	Constants
Units	one per input (includes the target's own past) + residual latent state (poles 0.8 and 0.95); up to 4 active
Response	bands [1], [2–3], [4–7], [8–15], [16–31] + peak lag (chosen at 200 samples) + current value; splits into halves
Challengers	RLS within episode, $P_0 = 10 \cdot I$; warm-up 250; 2 slots; one warm-up at a time
Evidence	loss $\min\{e^2/B^2, 1\}$, $B = 2 \cdot \max(\hat{\sigma}_S, 0.1 \cdot s_slow)$; $c = 1 + \max(\varepsilon, 0)$; $8\ \lambda = 0.9 \cdot 2^{-i}/c$; $\varepsilon = +0.002$ (add, swap, split), -0.002 (remove)
Multiplicity	$\alpha = 0.05$; $\gamma = 1/(2P)$ for the first $P = 2 \cdot \lvert\text{units}\rvert$, then $0.5 \cdot \gamma_k \propto 1/(k \log^2 k)$; level $\alpha \cdot \gamma \cdot (R+1) \le 0.5$
Episodes	scan after 100 samples; limit 5,000; wait 2,000; periodic removal every 2,000; decisions every 10 steps
Screening	prewhitened AR(1), weight 0.02, every 4 steps per part; thresholds 15.09 (χ^2_5) and 6.63 (χ^2_1)
Base of S	NLMS $\mu = 0.05$; input contract $\pm 8\sigma$ with a 31-step quarantine
Memory	NLMS $\mu = 0.05$; slow mean 0.01; profile 0.1; floors 10^{-3} and 10^{-4}; declared cycle(s); hold during gaps
Combination and output	$\lambda = 0.99$, $\eta = 0.5$, every 8 steps; envelope with rate 10^{-3} and margin 0.5×
Interval	$\alpha = 0.1$, $\gamma = 0.1$, every 8 steps (also on the final output)
Groups	causal correlation ≥ 0.95 reported with the accepted input (does not affect decisions)

Part	Parameter	Value	Note
M1	inputs	$m = \min(d, 5)$	largest absolute correlation (exponential means 0.01, past only)
M1	bands	[0], [1], [2–3], [4–7], [8–15]	means; + target at lags 1 and 2; $k = 3 + 5m$
M1	RLS	$\lambda = 0.999$; $\delta = 100$	every 8 learned targets; E only after k updates
M1	selection	after 100; every 5,000	restarts regression and combination
M1	clip	$L \pm 2 \cdot \max(\hat{\sigma}_L, \text{floor})$	floor: 0.1 × slow target scale
M1	combination	AdaHedge	squared error; 100-update shadow
M2	R	zero / seasonal / persistence	declared with the data
M2	H	AdaHedge	squared error, only with R defined
M2	W	one-way switch, hazard 1/(n+1)	Gaussian losses at the recent scale (0.01)
M2	Prod	$\eta \le ½$; $w_B = ½$	losses ÷ max(losses, 0.99·N)
M2	gate	$\varepsilon = 0.002$; $\alpha = 0.01$	audit only; recreated after each decision
M1, dormancy	sleep	$\omega < 10^{-3}$ for 2,000 targets	only after the shadow
M1, dormancy	asleep	E every 8; RLS every 64	output L' = L; AdaHedge frozen
M1, dormancy	wake	EMA(0.01) of $g > 0.05$	after 100 samples; AdaHedge restarts with a 100-update shadow
M1, dormancy	trial on reselection	sleeps again after 500 targets	unless ω reaches 10^{-3}
Core, removal	trigger	contribution < 0.5 × reference	reference: peak since acceptance, resets at each test; ≥ 2,000 steps between tests
Core, removal	evidence	new hypothesis per episode; $\varepsilon = -0.002$	level $\alpha \cdot \gamma_k \cdot (R\text{^rem} + 1)$, $\alpha = 0.05$, own budget
Core, removal	duration	up to 10,000; accepts only after 5,000	priority over the periodic test of the same unit
Core, report	equivalence matrix	one sample every 31 steps; weight 1e-3 × 31	applied in up to ⌈d/30⌉ rows per step; queries apply the pending ones

24. Pseudocode and reproducibility conditions

Simplified pseudocode of one step, sufficient to re-implement the evaluated configuration. Where text and pseudocode differ, the pseudocode together with the parameter table (§23) prevails.

Structural core

```
CONSTANTS  # the same for any series; nothing is tuned per series
  BANDS = [1], [2–3], [4–7], [8–15], [16–31]      POLES = 0.8, 0.95        L = 31
  M_MAX = 4   SLOTS = 2   WARM = 250   PEAK_N = 200   N_MIN = 100   T_MAX = 5000   COOL = 2000   REM_PERIOD = 2000
  DECIDE_EVERY = 10   SCREEN_EVERY = 4   SCREEN_W = 0.02   Q_UNIT = 15.09 (χ²₅, 99%)   Q_SPLIT = 6.63 (χ²₁, 99%)
  ε: add = swap = split = +0.002, remove = −0.002     α = 0.05     μ_S = 0.05     μ_M = 0.05     CLIP_X = 8
  λ_DMA = 0.99   η = 0.5   EVERY = 8   α_int = 0.10   γ_int = 0.10   SCALE_FLOOR = 0.1   OUT_MARGIN = 0.5   OUT_RHO = 1e-3

STATE  # fixed size, given the number of inputs d and the period(s) s, s2
  H[0..L+1][d+1]  input history (includes the target's own past, standardised)
  band_sum[5][d+1], seg_sum{(i,lo,hi)}   running sums of the bands and of active segments
  base[d+2]  active{unit → (segments, weights)}  r[2]  residual latent states
  M: buffer, w_M, G[s], m_slow        hyps{key → e-process}        slots[≤ 2]

STEP(x_t, y_t or gap, quarantine):
  # 1. read
  x̃ ← [x_t, std_causal(y_last)]                                  # EMA mean/variance 1e-4, start (0, 1)
  if any |x̃| > CLIP_X: quar_until ← t + L;   x̃ ← clip(x̃, ±CLIP_X);   shift H, band_sum, seg_sum
  # 2. predict
  ŷ_M ← M.predict()                                             # last value + w_M·ϕ (reversion, difference, seasonal, profile)
  ŷ_S ← base·[1, x̃] + Σ_u w_u · block(u)                         # block: means of the active segments
  ŷ ← ŷ_M + w_S (ŷ_S − ŷ_M)
  if n_obs ≥ 50: ŷ ← clip(ŷ, env_lo − 0.5·W, env_hi + 0.5·W), W = env_hi − env_lo;  if ŷ not finite: ŷ ← y_last
  for each experiment e: g_e ← ŷ_S + θ_e·z_e − (removed contribution)
  return ŷ, [ŷ − q̂, ŷ + q̂]
  # 3. observe and learn (only if y_t is observed)
  learn ← y_t observed ∧ ¬quarantine ∧ t > quar_until
  B ← 2·max(σ_r, SCALE_FLOOR·sd_slow(y));  ℓ(u) = min((y_t − u)²/B², 1)
  if learn:
    for each e: k_e += 1
       if k_e > WARM: e.hyp.observe(ℓ(ŷ_S) − ℓ(g_e) − ε_e)         # evidence
       if k_e ≤ PEAK_N: accumulate prewhitened correlation per lag (half the lags per step)
       if k_e = PEAK_N: k* ← argmax_k c_xy²/c_xx;  if k* > 1: insert x_i(t−k*) into the challenger
       RLS(θ_e, P_e; z_e, alvo = y_t − ŷ_S + removed contribution)       # P₀ = 10·I
    NLMS(base, w_u; res = y_t − ŷ_S, μ_S, norm = ‖[1,x̃]‖² + Σ‖block(u)‖²)
    screen_spread(res)                                          # each part on its own phase, every 4 steps
  D ← λ_DMA·D + ((y_t − ŷ_S)² − (y_t − ŷ_M)²)/σ²;   every EVERY: w_S ← 1/(1 + exp(η·D))
  interval: miss ← |y_t − ŷ| > q̂;   every EVERY: q̂ ← max(0, q̂ + γ_int·σ·Σ(miss − α_int))
  M.update(y_t if observed; otherwise the last observed value (hold), without learning)
  env ← envelope of observed values (max/min relaxing to the mean, rate OUT_RHO)
  r ← POLES·r + (1 − POLES)·(y_t − ŷ)/σ
  if t mod DECIDE_EVERY = 0: DECIDE()

DECIDE():
  for each e: if log E(e.hyp) ≥ log(1/α_hyp): ACCEPT(e)
                else if n_e ≥ T_MAX or (n_e ≥ N_MIN and Σ_episode d ≤ 0): close (the evidence stays with the hypothesis)
  close experiments about units that just changed;  FILL()

FILL():  while a slot is free:
  1. active unit without a removal test for ≥ REM_PERIOD → open remove
  2. if some challenger is warming up: stop        # one warm-up at a time
  3. u* ← argmax stat(u) among inactive, free units out of cool-down;  s* ← argmax stat(splits)
  4. pick the larger of stat(u*)/Q_UNIT and stat(s*)/Q_SPLIT (if ≥ 1):
       split → split;  |active| < M_MAX → add;  else → swap (with the unit of smallest contribution)
  # a new hypothesis n receives a fixed level: α_n = min(α·γ_n·(R+1), 0.5),
  # γ_n = 1/(2P) for n ≤ P = 2·|units|, then 0.5·C/((n−P)·log²(n−P+1)); R = changes accepted so far

e-process(ε):  c = 1 + max(ε, 0);  λ_j = 0.9·2^(−j)/c, j = 0..7;  ψ_j = (−log(1 − cλ_j) − cλ_j)/c²
  observe(d): γ ← clip(past mean of d, −c, 0);  S += d;  V += (d − γ)²
  log E = log( (1/8)·Σ_j exp(λ_j·S − ψ_j·V) )

ACCEPT(e):  add/swap → active[i] ← (BANDS ∪ {[k*]}, θ_e[:-1]);  base[i] += θ_e[-1];  remove → delete active[u]
            split → replace band [lo,hi] by its two halves with the challenger weights;  R += 1
```

M1 and M2 (on top of the core forecast L)

```
M1 and M2 CONSTANTS   M_MAX = 5   BANDS_E = [0], [1], [2–3], [4–7], [8–15]   λ_RLS = 0.999   δ_RLS = 100
              EVERY = 8   CLIP_K = 2   N_MIN = 100   T_MAX = 5000   LAM = 0.99   SCALE_FLOOR = 0.1   ε_gate = 0.002   α_gate = 0.01

PREDICT(x_t):
  L ← core.predict(x_t)                              # core pseudocode above
  # M1: precision expert
  x ← x_t with missing values replaced by the last value seen
  z ← [1, ỹ_(t−1), ỹ_(t−2), BANDS_E means of the m = min(d, M_MAX) chosen inputs, standardised]
  E ← clip(w·z, L ± CLIP_K·max(σ̂L, floor))      if RLS determined (n_RLS ≥ k = 3 + 5m)
  L' ← L if E undefined or AdaHedge₁.n < N_MIN; else (1 − ω)·L + ω·E      # shadow
  # M2: start from the trivial reference
  R ← 0 | y_last | y_(t−s)                                  # zero, persistence or seasonal naive, declared
  if R undefined: ŷ ← L'
  else: H ← w_H·[R, L'];  W ← w_W·[R, L'];  s ← η·w_A / (η·w_A + w_B/2);  ŷ ← s·W + (1 − s)·H
  return ŷ, [ŷ − q̂_o, ŷ + q̂_o]

OBSERVE(y_t, quarantine):
  core.observe(y_t, quarantine)
  if y_t observed:
    AdaHedge₁.update((y_t − L)², (y_t − E)²)                  # squared error, no clipping
    learn ← ¬quarantine ∧ inputs within the contract
    if learn: every EVERY targets: screening (exponential means 1 − LAM) and RLS(z, y_t)
             selection of the m inputs after N_MIN targets and every T_MAX; if it changes: RLS and AdaHedge₁ restart
    if ¬quarantine ∧ R defined:
      AdaHedge_H.update((y_t − R)², (y_t − L')²)
      s² ← max(min(σ²_R, σ²_L'), floor²);  Switch_W.update((y_t − R)²/2s², (y_t − L')²/2s²)   # log-scale fallback against 0/0
      N ← max(LAM·N, (y_t − H)², (y_t − W)²);  Prod.update((y_t − H)²/N, (y_t − W)²/N)
      gate.observe(...)                                       # audit: logs events, does not decide the output
    output interval: miss ← |y_t − ŷ| > q̂_o;  every EVERY: q̂_o ← max(0, q̂_o + 0.1·σ̂_o·Σ(miss − 0.1))
```

Mechanisms of v0.54

```
DORMANT M1 # replaces M1 above while asleep
  if active ∧ after the shadow ∧ ω < 10⁻³ for D consecutive targets: sleep          # D = 2,000; D = 500 in trial
  if asleep: L' ← L
    every 8 targets: E computed; g ← ((y − L)² − (y − E)²)/(y − L)²;  ḡ ← ḡ + 0.01·(g − ḡ)
    every 64 targets: RLS(z, y)
    if ḡ > 0.05 ∧ samples ≥ 100: wake; AdaHedge₁ restarts with shadow
    if the input selection changed: wake in trial

TRIGGERED REMOVAL # at each core decision, before the periodic schedule
  for each active unit u: ref_u ← max(ref_u, contrib_u)
  if contrib_u < 0.5·ref_u ∧ ≥ 2,000 steps since u's last triggered test ∧ u free or only in a periodic removal test:
    close u's periodic test (the persistent hypothesis is kept)
    open an episode with a new hypothesis; level α·γ_k·(R^rem + 1);  ref_u ← contrib_u
    accept if log E ≥ log(1/α_k) ∧ evidence steps ≥ 5.000;  close at 10,000 or by the scan rule

EQUIVALENCE REPORT
  every 31 steps: apply the pending rows; m ← m + 0.031·(x − m);  xc ← x − m
  every step: apply xc to up to ⌈d/30⌉ rows i: C[i, j≥i] ← C[i, j≥i] + 0.031·(xc_i·xc_j − C[i, j≥i]); mirror
  query: apply the pending rows; group of u = inputs with |correlation| ≥ 0.95
```

Reproducibility conditions

- **Inputs:** standardise each input causally before the core: z = (x − m)/√(v + 10^{-6}), with exponential m and v at rate 10^{-4} from (0, 1), updated *after* standardising the current value; missing values become 0 after standardisation (in M1, the last value seen).
- **Target:** passed in original units; missing values are passed as a gap (no learning); an external quarantine flag suspends learning.
- **Metadata:** seasonal period s (24 for hourly data; none if there is no cycle), optionally a second cycle s_2 (168 for buildings), and the trivial reference (zero, persistence or seasonal; if not declared: seasonal if there is a cycle, otherwise persistence).
- **Determinism:** the algorithm uses no random numbers; given the same input sequence, the output is deterministic.
- **Availability:** the promoted code (Python), the C port of the core, the pre-registrations, the analysis scripts, the per-series forecasts and the SHA-256 manifests of every frozen version are in the public repository github.com/Iquitim/lebre-research (code of the complete LEBRE, with errata 01, at commit f9906c4; the v0.54 final evaluation ran with commit cfe72ce, with identical forecasts); the development and validations are in the public test bench github.com/Iquitim/lebre-lab (this document read its results at commit 7acea5f).

- **Environment used:** Python 3.11, numpy 2.2.5, pandas 2.2.3 (later versions changed read-only array behaviour), statsmodels 0.15.0; Chronos-2 with the chronos 2.3.2 package; C99 with Arm GCC 15.2 (−O2, single-precision FPU, no FMA contraction); Renode 1.17.

25. References

1. Adamskiy, D., Koolen, W. M., Chernov, A., Vovk, V. (2016). A closer look at adaptive regret. *Journal of Machine Learning Research* 17(23), 1–21.
2. Alquier, P. (2021). Non-exponentially weighted aggregation: regret bounds for unbounded loss functions. *ICML*.
3. Amoukou, S. I., Mishra, S., Veloso, M. (2026). Correcting split selection in online decision trees via anytime-valid inference. *arXiv:2605.31239*.
4. Ansari, A. F. et al. (2025). Chronos-2: from univariate to universal forecasting. *arXiv preprint*.
5. Box, G. E. P., Jenkins, G. M. (1970). *Time Series Analysis: Forecasting and Control*. Holden-Day.
6. Campbell, J. Y., Yogo, M. (2006). Efficient tests of stock return predictability. *Journal of Financial Economics* 81(1), 27–60.
7. Cesa-Bianchi, N., Gaillard, P., Lugosi, G., Stoltz, G. (2012). Mirror descent meets fixed share (and feels no regret). *NeurIPS 25*.
8. Cesa-Bianchi, N., Mansour, Y., Stoltz, G. (2007). Improved second-order bounds for prediction with expert advice. *Machine Learning* 66, 321–352.
9. Chagas, V. B. P. et al. (2020). CAMELS-BR: hydrometeorological time series and landscape attributes for 897 catchments in Brazil. *Earth System Science Data* 12, 2075–2096.
10. Chen, S. (2017). Beijing Multi-Site Air Quality. *UCI Machine Learning Repository*, doi:10.24432/C5RK5G.
11. Choe, Y. J., Ramdas, A. (2024). Comparing sequential forecasters. *Operations Research* 72(4), 1368–1387.
12. Choi, S. (2026). Anytime-valid evidence for prespecified predictive corrections. *arXiv:2608.08174*.
13. Clark, T. E., West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics* 138(1), 291–311.
14. de Heide, R. (2026). Dynamic e-closure for online hypotheses with any-time-valid evidence: closure principles and projective mergers. *arXiv:2608.09927*.
15. de Rooij, S., van Erven, T., Grünwald, P. D., Koolen, W. M. (2014). Follow the leader if you can, hedge if you must. *Journal of Machine Learning Research* 15, 1281–1316.
16. Ekambaram, V. et al. (2024). Tiny Time Mixers (TTMs): fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series. *NeurIPS 37*.
17. Even-Dar, E., Kearns, M., Mansour, Y., Wortman, J. (2008). Regret to the best vs. regret to the average. *Machine Learning* 72, 21–37.
18. Fan, X., Grama, I., Liu, Q. (2015). Exponential inequalities for martingales with applications. *Electronic Journal of Probability* 20, 1–22.
19. Fernández-Barrios, R., Pastor-López, I., Pikatza-Huerga, A., García Bringas, P. (2026). Candidate comparability before promotion: conditional validation in adaptive network intrusion detection. *arXiv:2609.04388*.
20. Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica* 74(6), 1545–1578.
21. Gibbs, I., Candès, E. J. (2021). Adaptive conformal inference under distribution shift. *NeurIPS 34*.
22. Grünwald, P., de Heide, R., Koolen, W. M. (2024). Safe testing. *Journal of the Royal Statistical Society B* 86(5), 1091–1128.
23. Han, W., Qu, L. (2026). When the martingale never stops firing: anytime-valid gating on real forecast streams. *arXiv:2608.30502*.
24. Haykin, S. (2002). *Adaptive Filter Theory*, 4th ed. Prentice Hall.
25. Hazan, E., Kale, S. (2010). Extracting certainty from uncertainty: regret bounded by variation in costs. *Machine Learning* 80, 165–188.
26. Howard, S. R., Ramdas, A., McAuliffe, J., Sekhon, J. (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics* 49(2), 1055–1080.
27. Hyndman, R. J., Koehler, A. B., Ord, J. K., Snyder, R. D. (2008). *Forecasting with Exponential Smoothing: The State Space Approach*. Springer.
28. Javanmard, A., Montanari, A. (2018). Online rules for control of false discovery rate and false discovery exceedance. *Annals of Statistics* 46(2), 526–554.
29. Jha, A. (2026). Distributional Granger causality: identification, sequential inference, and adaptive testing. *arXiv:2606.22230*.
30. Koolen, W. M. (2013). The Pareto regret frontier. *NeurIPS 26*.
31. Lara Pinal, E. M., Das, A., Schlüter, S. (2026). A low-cost IoT device for environmental monitoring and embedded solar forecasting with on-device incremental learning. *arXiv:2608.14698*.
32. Lin, S. et al. (2024). SparseTSF: modeling long-term time series forecasting with 1k parameters. *ICML*.
33. Ljung, L. (1999). *System Identification: Theory for the User*, 2nd ed. Prentice Hall.
34. Mallat, S. (1989). A theory for multiresolution signal decomposition: the wavelet representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 11(7), 674–693.
35. Meinshausen, N. (2008). Hierarchical testing of variable importance. *Biometrika* 95(2), 265–278.
36. Mhammedi, Z., Koolen, W. M., van Erven, T. (2019). Lipschitz adaptivity with multiple learning rates in online learning. *COLT*.
37. Miller, C. et al. (2020). The Building Data Genome Project 2, energy meter data from the ASHRAE Great Energy Predictor III competition. *Scientific Data* 7, 368.
38. Moulin, A., Esposito, E., van der Hoeven, D. (2025). When lower-order terms dominate: adaptive expert algorithms for heavy-tailed losses. *arXiv:2506.01722*.
39. Orabona, F., Pál, D. (2018). Scale-free online learning. *Theoretical Computer Science* 716, 50–69.
40. Raftery, A. E., Kárný, M., Ettler, P. (2010). Online prediction under model uncertainty via dynamic model averaging. *Technometrics* 52(1), 52–66.
41. Ramdas, A., Grünwald, P., Vovk, V., Shafer, G. (2023). Game-theoretic statistics and safe anytime-valid inference. *Statistical Science* 38(4), 576–601.
42. Sani, A., Neu, G., Lazaric, A. (2014). Exploiting easy data in online optimization. *NeurIPS 27*.
43. Stambaugh, R. F. (1999). Predictive regressions. *Journal of Financial Economics* 54(3), 375–421.
44. van Erven, T., Grünwald, P., de Rooij, S. (2012). Catching up faster by switching sooner: a predictive approach to adaptive estimation with an application to the AIC–BIC dilemma. *Journal of the Royal Statistical Society B* 74(3), 361–417.
45. V'yugin, V., Trunov, V. (2019). Online aggregation of unbounded losses using shifting experts with confidence. *Machine Learning*; arXiv:1808.00741.
46. Wang, H., Ramdas, A. (2023). Catoni-style confidence sequences for heavy-tailed mean estimation. *Stochastic Processes and their Applications*.
47. Waudby-Smith, I., Ramdas, A. (2024). Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society B* 86(1), 1–27.
48. Widrow, B., Stearns, S. D. (1985). *Adaptive Signal Processing*. Prentice Hall.
49. Xu, Z., Ramdas, A. (2024). Online multiple testing with e-values. *AISTATS*.
50. Xu, Z., Zeng, A., Xu, Q. (2024). FITS: modeling time series with 10k parameters. *ICLR*.

51. Yao, Z., Gang, B., Sun, W. (2025). Safe, always-valid alpha-investing rules for doubly sequential online inference. *arXiv:2512.12244*.
52. Yekutieli, D. (2008). Hierarchical false discovery rate-controlling methodology. *Journal of the American Statistical Association* 103(481), 309–316.
53. Zeng, A., Chen, M., Zhang, L., Xu, Q. (2023). Are transformers effective for time series forecasting? (DLinear, NLinear). *AAAI*.

26. Revision history

- **Revision 0** (2026-10-14): first edition of v0.54, self-contained — it describes the whole LEBRE (core with triggered removal and cheap equivalence report, M1 with dormancy, and M2) without assuming the reading of any other document; it includes the v0.54 final evaluation, the stress audit and errata 01.